Trigger efficiency for the diphoton low mass trigger in 2024

HLT_Diphoton30_18_R9IdL_AND_HE_AND_IsoCaloId

Santanu Mahata (TIFR, Mumbai)
on behalf of the low-mass H→*γγ* team (HIG-25-016)













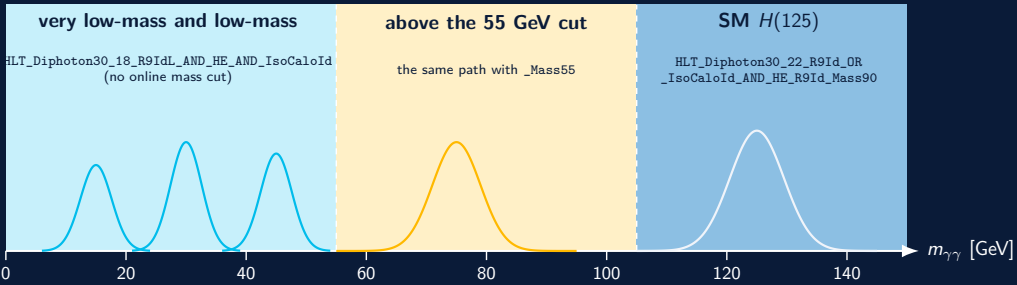








Choosing HLT according to the physics



- An online $m_{\gamma\gamma}$ requirement cuts away exactly the region a search below it needs — so the **no-mass-cut** path is the one that works for the very low-mass and low-mass searches, and it is the path measured here.
- Above the cut the _Mass55 variant is usable, and the SM $H \to \gamma\gamma$ analyses run on the _Mass90 path.

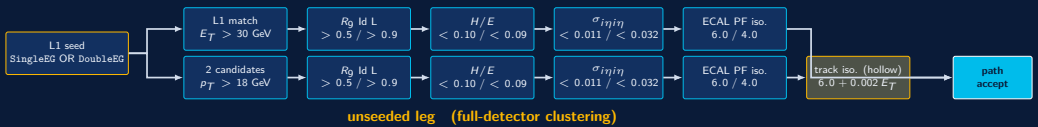
Peak positions and widths are illustrative, not measured line shapes.

The HLT path: reading the name, and the filter sequence

`HLT_Diphoton30_18_R9IdL_AND_HE_AND_IsoCaloId`

`HLT_`	runs on the full event, after the L1 accept	`R9IdL`	R_9 identification, loose working point
`Diphoton`	two photon candidates: one seeded, one unseeded leg	`HE`	hadronic-over-electromagnetic filter
`30_18`	the two p_T thresholds in GeV	`IsoCaloId`	shower shape and calorimeter isolation

No `_Mass` suffix — this path applies no online $m_{\gamma\gamma}$ cut, which is why it is the one usable below 55 GeV.
L1 seed (`hltL1sSingleAndDoubleEGor`) — the OR of the single- and double-EG seeds.



Both legs must fire, so they are measured **separately**; the **hollow track isolation** exists only on the unseeded leg. EB / EE thresholds shown.

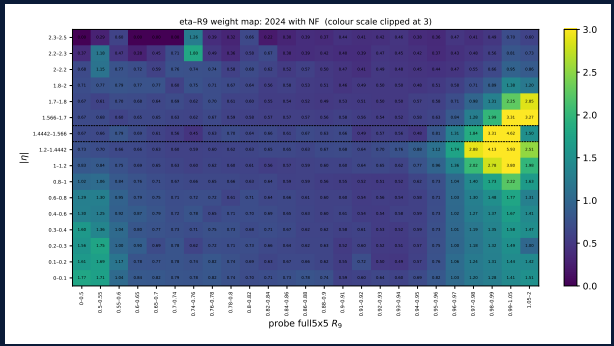
Offline photon preselection for 2024 (same as 2023), and the binning

Category	H/E	$\sigma_{i\eta i\eta}$	R_9	`Rho_PfPhoIso03`	`TrackerIso03`
Barrel, $R_9 < 0.85$	< 0.08	< 0.011	> 0.5	< 3.0	< 6.0
Barrel, $R_9 > 0.85$	< 0.08	< 0.011	> 0.5	< 3.0	< 6.0
Endcap, $R_9 > 0.9$	< 0.08	< 0.032	> 0.9	< 3.0	< 6.0

Leg	Bins	Probe E_T bin edges [GeV]
seeded (30 GeV)	9	0, 35, 37, 40, 45, 50, 60, 70, 90, ∞
unseeded (18 GeV)	12	0, 21, 25, 28, 31, 35, 40, 45, 50, 60, 70, 90, ∞
5 R_9/η categories on each leg:	EB 0.50–0.55, EB 0.55–0.85, EB > 0.85, EE 0.90–0.93, EE > 0.93	

Why an η–R_9 reweighting

- The shower shapes of photons and electrons are different.
- The efficiency is derived with electrons ($Z \to ee$), so it has to be corrected for the difference in shower shape.
- The correction is applied as a weight in bins of $|\eta|$ and R_9.



η–R_9 weight map, 2024 with NF.

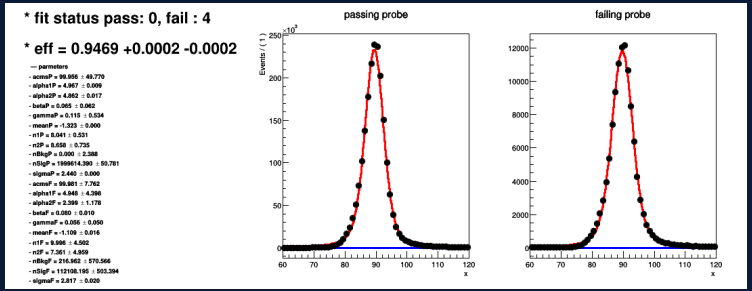
Tag-and-probe method

- **Sample:** $Z \to ee$ events in 2024 data; electrons are used as photon proxies since the online filters are photon-like (no pixel-veto at HLT here).
- **Tag:** tightly identified, trigger-matched electron. **Probe:** the other leg, unbiased with respect to the filter under test.
- Efficiency per (leg, ECAL region, R_9 bin, probe-p_T bin) from a **fit** of the tag–probe invariant-mass distribution in the passing and failing categories. It is the **efficiency of the probe** — N_{pass} and N_{fail} count **probes** that do and do not pass the filter under test:

$$\varepsilon_{\text{probe}} = \frac{N_{\text{pass}}^{\text{probe}}}{N_{\text{pass}}^{\text{probe}} + N_{\text{fail}}^{\text{probe}}}$$

- **Tag trigger:** `HLT_Ele30_WPTight_Gsf`, tag matched to the filter of this trigger.
- **p_T cuts:** tag $p_T > 35$ GeV; probe $p_T > 30$ GeV (seeded leg) and $p_T > 18$ GeV (unseeded leg).
- **Signal model:** Breit–Wigner $\otimes$ double-sided Crystal Ball (`bwDscb`).
- **Background:** **RooCMSShape** (nominal) — an **Exponential** fit is also run and is used as the fit-model variation.
- An η–R_9 **reweighting** and a **scale correction** are applied to the sample before fitting.

Example tag-and-probe mass fits



seeded, EB $0.55 < R_9 < 0.85$, $40 < p_{\mathrm{T}} < 45$ GeV

Passing and failing categories fitted; bwDscb signal, RooCMSShape background.

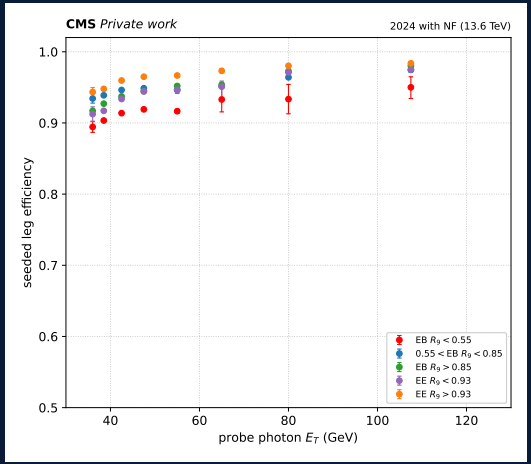
2024 dataset

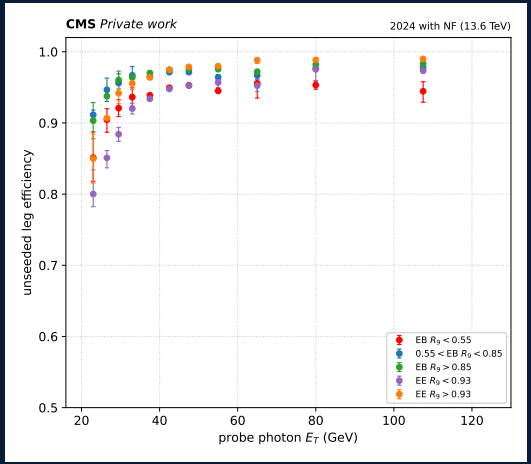
Data: `/EGamma0/*` and `/EGamma1/*` MINIAOD, **PromptReco**, eras **C–I**:

`/EGamma0/Run2024C-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024C-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024D-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024D-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024E-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024E-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024E-PromptReco-v2/MINIAOD`	`/EGamma1/Run2024E-PromptReco-v2/MINIAOD`
`/EGamma0/Run2024F-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024F-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024G-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024G-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024H-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024H-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024I-PromptReco-v1/MINIAOD`	`/EGamma1/Run2024I-PromptReco-v1/MINIAOD`
`/EGamma0/Run2024I-PromptReco-v2/MINIAOD`	`/EGamma1/Run2024I-PromptReco-v2/MINIAOD`

- Photon energy scale & smearing applied from the Summer24 NanoAODv15 EGM `photonSS_EtDependent` correction set.
- 2024 is treated as a **single era** — no pre/post split, unlike 2022 (preEE/postEE) and 2023 (pre/postBPix).

2024 efficiency with NF — seeded and unseeded legs



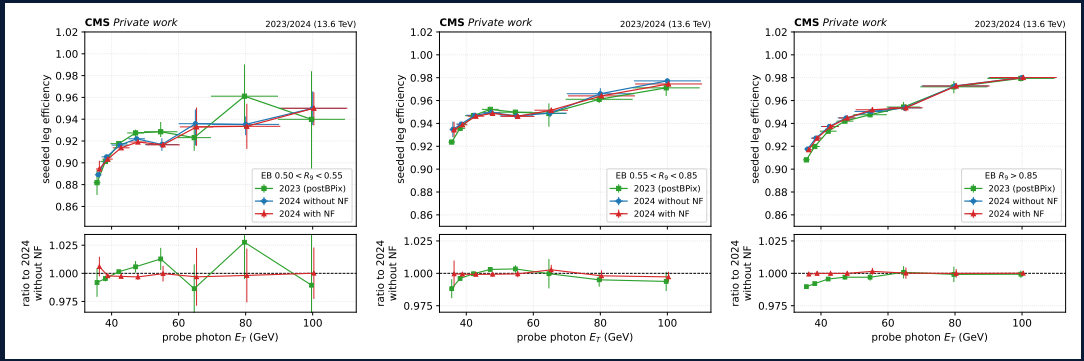


Data efficiency vs probe E_T, 2024 with NF; the five R_9/η categories overlaid. Nominal fit. Error bars: stat $\oplus$ syst.

Inspecting the effect of normalising flow corrections

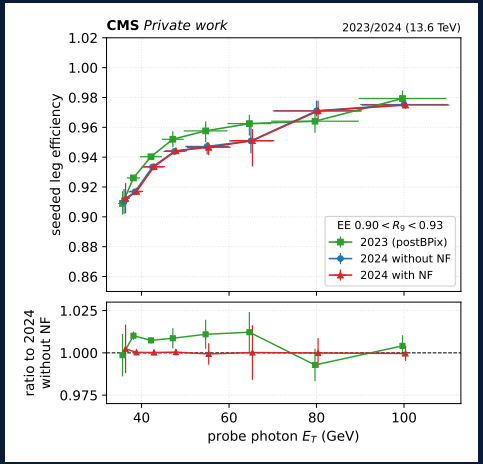
Set	η–R_9 reweighting map
2023 (postBPix)	with flow-corrected η–R_9 reweighting map
2024 without NF	without flow-corrected η–R_9 reweighting map
2024 with NF	with flow-corrected η–R_9 reweighting map

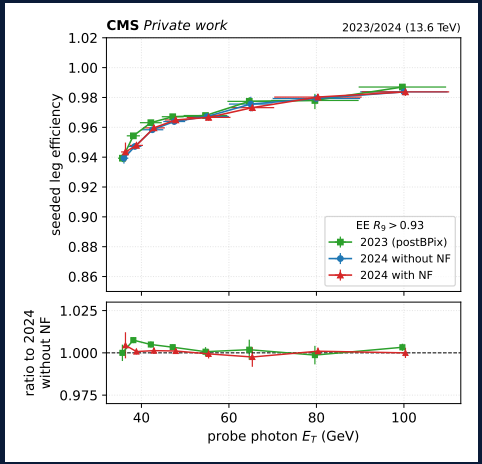
2023 vs 2024 without NF vs 2024 with NF — seeded leg, EB bins



Error bars: stat ⊕ syst. Lower pad: ratio to 2024 without NF, the two uncertainties added in quadrature.

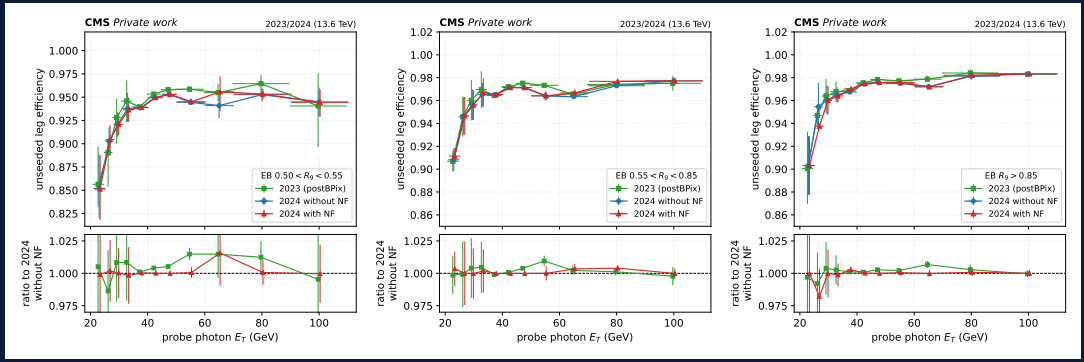
2023 vs 2024 without NF vs 2024 with NF — seeded leg, EE bins





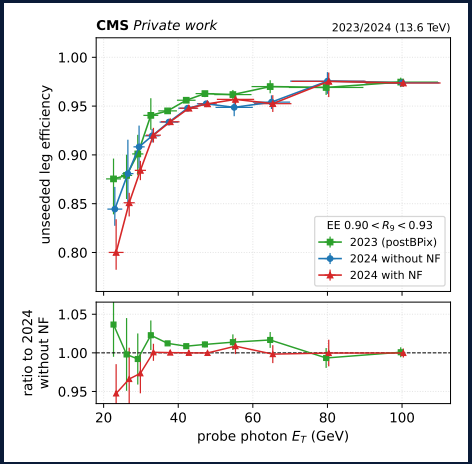
Error bars: stat ⊕ syst. Lower pad: ratio to 2024 without NF, the two uncertainties added in quadrature.

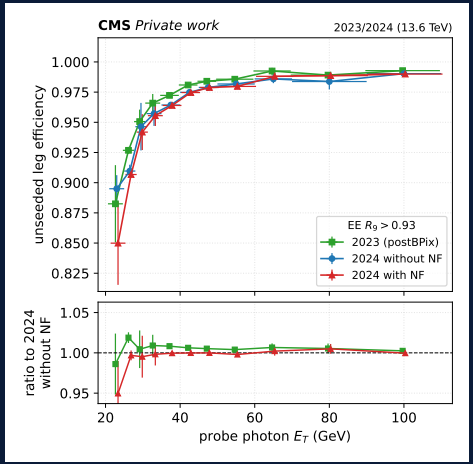
2023 vs 2024 without NF vs 2024 with NF — unseeded leg, EB bins



Error bars: stat ⊕ syst. Lower pad: ratio to 2024 without NF, the two uncertainties added in quadrature.

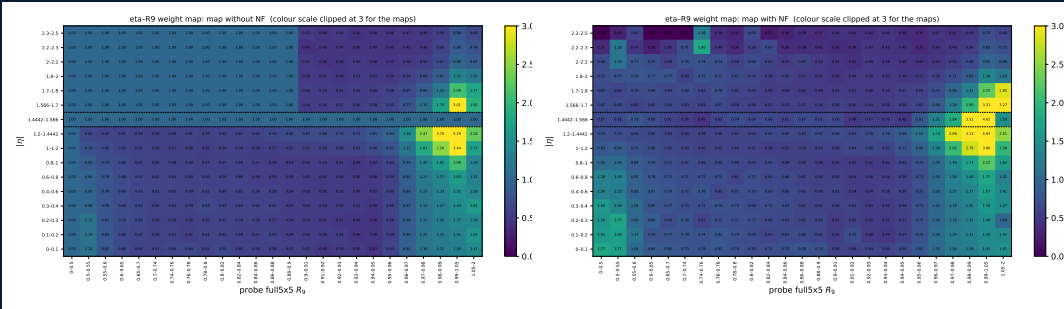
2023 vs 2024 without NF vs 2024 with NF — unseeded leg, EE bins





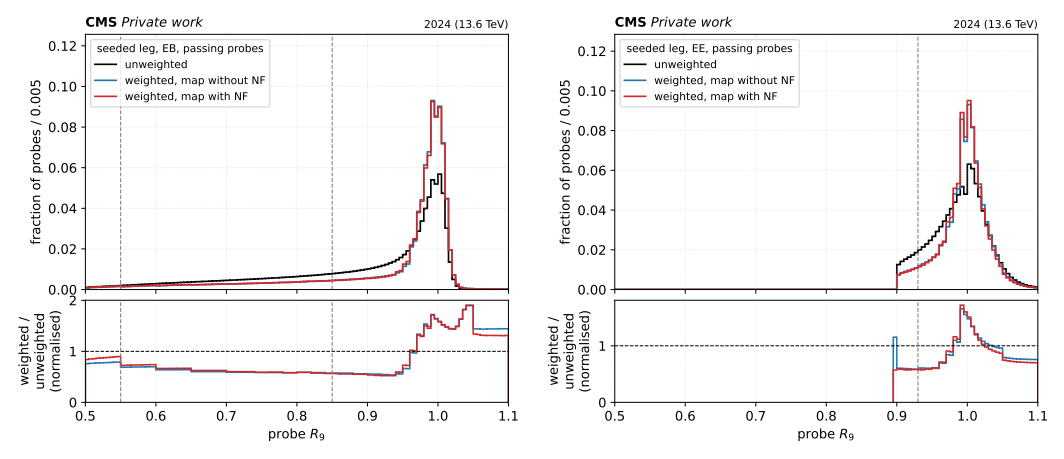
Error bars: stat ⊕ syst. Lower pad: ratio to 2024 without NF, the two uncertainties added in quadrature.

Map without NF vs map with NF



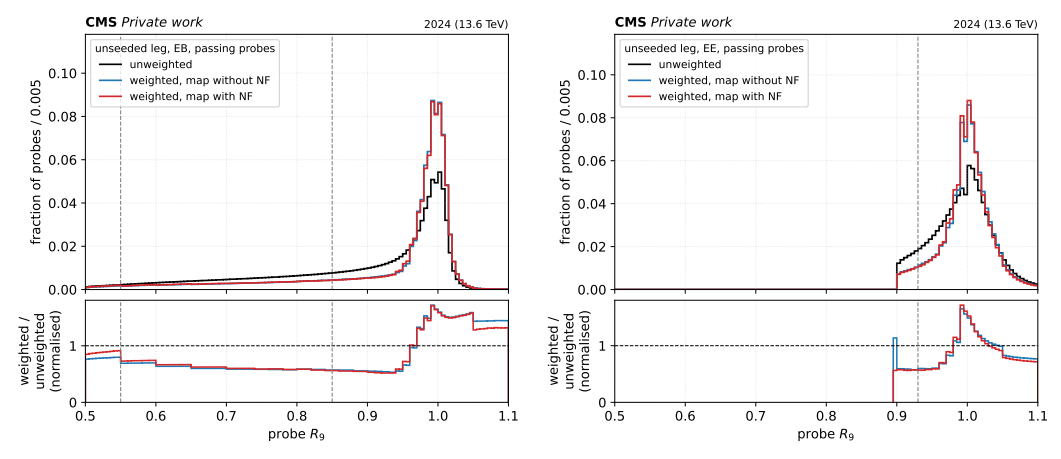
- Without NF: Hgg / DY MC, density-normalised, empty-DY cell → 1 (`.meta` file). Range 0.42–5.29; 118 cells exactly 1.
- With NF: no metadata file; recorded as "used as supplied" at weight replacement. Range 0–5.93; 4 cells exactly 0.

Probe R_9 before and after the η–R_9 reweighting — seeded leg



Passing probes of the normal-set fits, $60 < m_{ee} < 120$ GeV. Black: unweighted. Blue: weighted with the map without NF. Red: weighted with the map with NF. Each curve normalised to its total. Dashed lines: fit-category boundaries.

Probe R_9 before and after the η–R_9 reweighting — unseeded leg



Passing probes of the normal-set fits, $60 < m_{ee} < 120$ GeV. Black: unweighted. Blue: weighted with the map without NF. Red: weighted with the map with NF. Each curve normalised to its total. Dashed lines: fit-category boundaries.

Summary

- 2024 HLT efficiencies of the seeded and unseeded legs derived with tag-and-probe on $Z \to ee$, using the η–R_9 reweighting map without and with the normalising-flow (NF) correction.
- Efficiencies compared as a function of probe E_{T}: 2023, 2024 without NF and 2024 with NF.
- The two maps compared cell by cell; probe R_9 distributions shown before and after reweighting for each map.
- Effect of NF on the efficiencies is bin dependent:
  - barrel: negligible;
  - endcap: significantly larger than in any other bin, especially around the turn-on in probe p_{T}.
- Status: the 2023 trigger efficiencies are blessed by EGM; the 2024 efficiencies (without NF) have been presented to EGM and are waiting for approval.

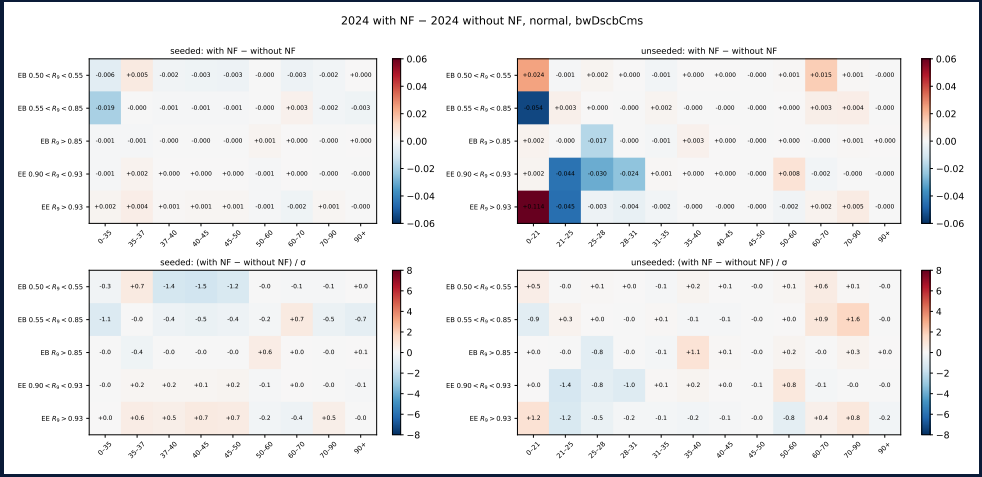
Backup

The HLT filter chain — online cuts (2024 sequence)

Filter	Variable	EB	EE
E_T (seeded)	`HLTEgammaEtFilter`	> 30 GeV	> 30 GeV
p_T (unseeded)	`HLT1Photon`	> 18 GeV	> 18 GeV
R_9 Id "L"	`r95x5`	> 0.5	> 0.9
H/E	`hltEgammaHoverE`	$< 0.10 \cdot E$	$< 0.09 \cdot E$
Cluster shape	$\sigma_{i\eta i\eta}$ (5×5)	< 0.011	< 0.032
ECAL PF-cluster iso.	`EcalPFClusterIso`	$< 6.0 + 0.012\, E_T$	$< 4.0 + 0.012\, E_T$
Track iso. **(unseeded only)**	hollow track iso.	$< 6.0 + 0.002\, E_T$	$< 6.0 + 0.002\, E_T$

Thresholds read directly from the 2024 filter-sequence dump (`thrRegular*` / `thrOverE*` of each `HLTEgammaGeneric[Quadratic]Filter`).

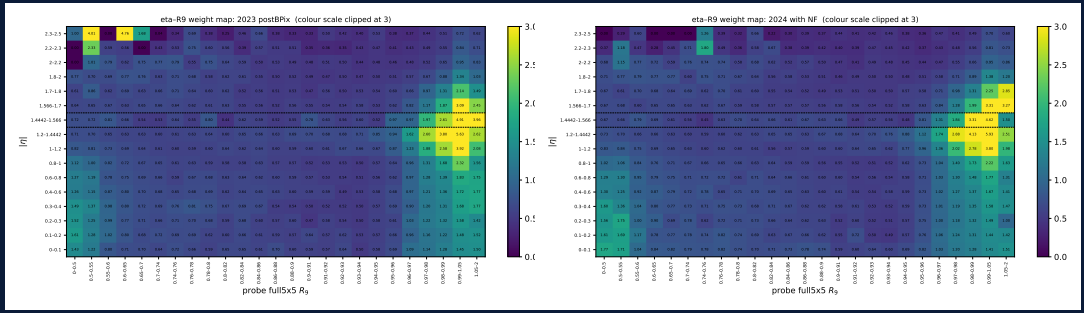
Where the 2024 efficiency moved: with NF − without NF, every bin



Top: absolute difference. Bottom: difference / ($\sigma_{\text{without NF}} \oplus \sigma_{\text{with NF}}$), σ = stat $\oplus$ syst; both fits use the same events, so this treats correlated numbers as independent.

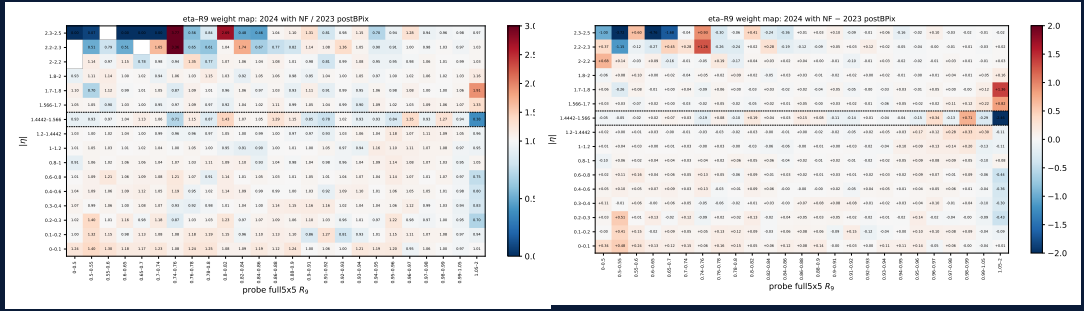
Mean $|\Delta| = 0.0048$ over 105 bins. 10 bins with $|\Delta| > 0.01$: 9 unseeded, 8 of them with $E_{\text{T}} < 31$ GeV. Largest $|\Delta|/\sigma$: 1.65.

η–R_9 map, 2023 postBPix vs 2024 with NF



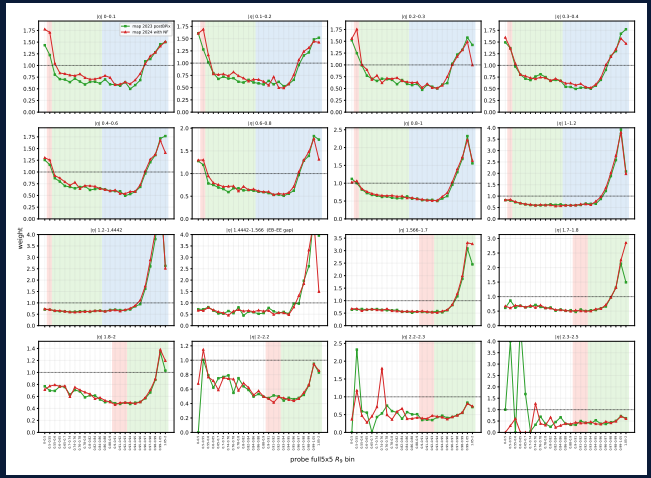
Same 25×16 binning. Range: 2023 postBPix 0.00–5.63, 2024 with NF 0.00–5.93. Cells exactly 0: 4 and 4, 0 in common.

2024 with NF vs 2023 postBPix, cell by cell



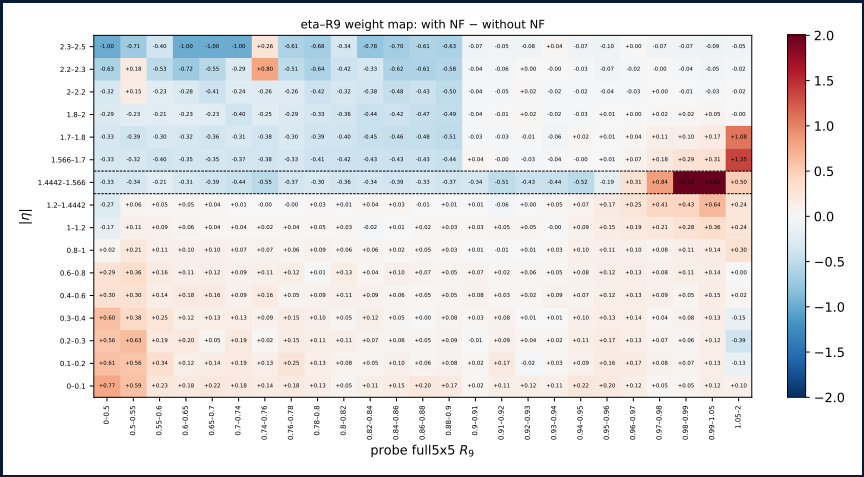
Left: ratio 2024 / 2023 (empty where the 2023 cell is 0). Right: difference 2024 − 2023. In the 282 cells of the analysis categories: ratio 0.70–1.91, mean |difference| 0.071, largest 1.36.

Map slices, 2023 postBPix vs 2024 with NF



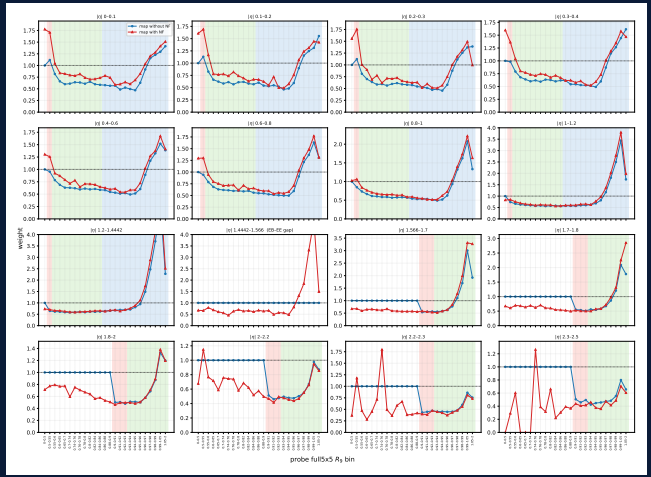
Weight vs R_9 bin in every $|\eta|$ row. Shaded: the R_9 windows of the analysis categories in EB and EE rows.

With NF − without NF, cell by cell



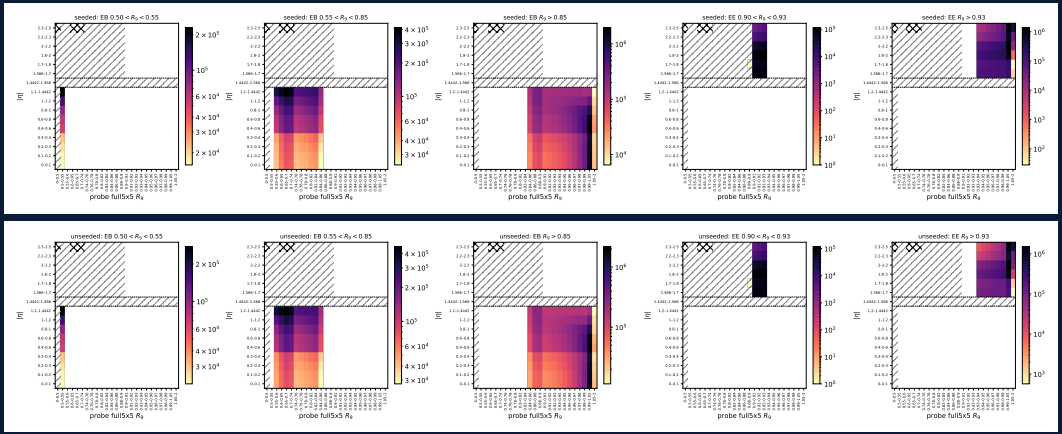
Drawn in bin-index space (R_9 bins are 0.5, 0.05, 0.02 and 0.01 wide); dashed lines bound the EB–EE gap row. with NF / without NF ranges 0 to 4.62.

Map slices: weight vs R_9 in every $|\eta|$ row



Shaded: the R_9 windows of the analysis categories in EB and EE rows.

Which cells the probes actually use

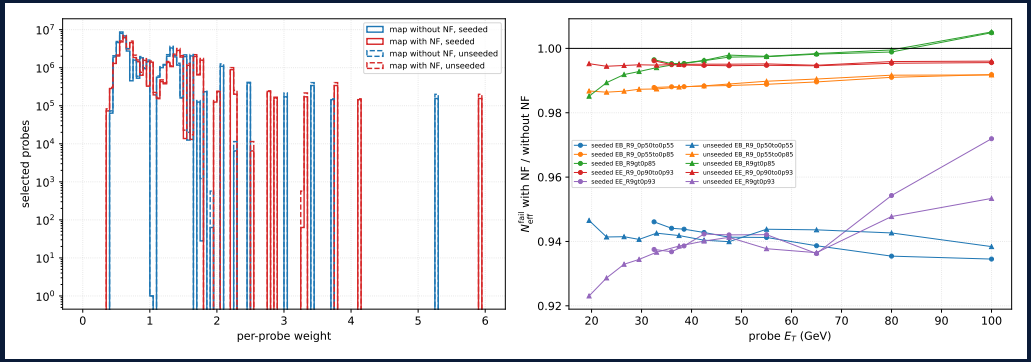


Selected probes per map cell (normal set, pass+fail, $60 < m < 120$ GeV), top seeded, bottom unseeded. Hatched: map without NF = 1 (//), map with NF = 0 (xx). Selected probes in either kind of cell: 0 in every category.

Backup

Do the weights explain the shifts?

Per-probe weights and effective statistics



Left: per-probe weight distribution of all selected probes. Right: effective number of failing probes $N_{\text{eff}} = (\sum w)^2 / \sum w^2$, with NF / without NF, per bin: 0.92–1.01.

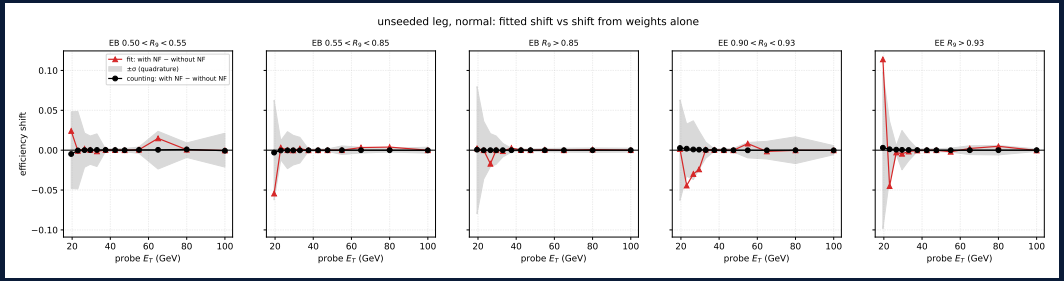
Method: what the weights alone do

- Both weights are a pure lookup in the map: every selected probe's stored weight equals its cell value (0 mismatches in $2\times159.9\,\mathrm{M}$ entries, without and with NF).
- So, per fitter bin, count probes per map cell once, separately for pass and fail. Any map then gives

$$P = \sum_c N_c^{\mathrm{pass}}\, w_c, \qquad F = \sum_c N_c^{\mathrm{fail}}\, w_c, \qquad \varepsilon_{\mathrm{count}} = \frac{P}{P+F}.$$

- Cuts, flags and E_{T} bins parsed from the 20 settings files; $60 < m < 120$ GeV. Rebuilt yields equal the sums of stored weights to 1.4×10^{-8} (relative).
- $\varepsilon_{\mathrm{count}}$ includes background — it is **not** the fitted efficiency. $\Delta\varepsilon_{\mathrm{count}} = \varepsilon_{\mathrm{count}}^{\mathrm{with\,NF}} - \varepsilon_{\mathrm{count}}^{\mathrm{without\,NF}}$ is the change from the map alone, with no fit.

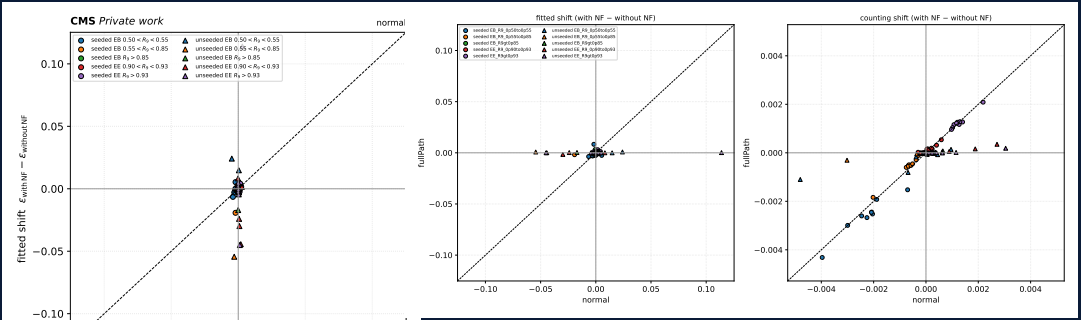
Fitted shift vs shift from the weights alone — unseeded



Red: fitted with NF − without NF. Black: $\Delta\varepsilon_{\mathrm{count}}$. Grey: $\pm(\sigma_{\mathrm{without\,NF}} \oplus \sigma_{\mathrm{with\,NF}})$, stat $\oplus$ syst.

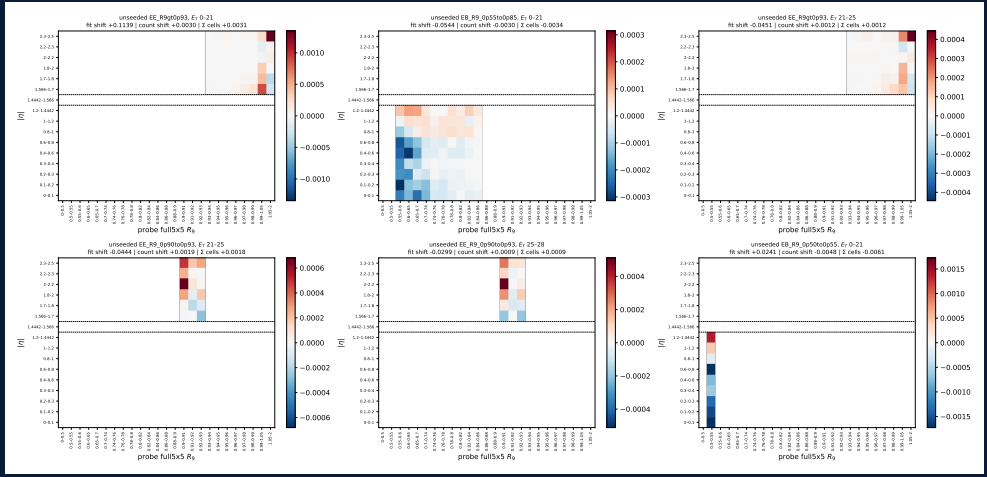
Largest $|\Delta\varepsilon_{\mathrm{count}}|$ in any bin: 0.0048 (normal), 0.0043 (fullPath). Largest fitted $|\Delta|$: 0.114.

Every bin: fitted shift vs counting shift; normal vs fullPath



Correlation of fitted and counting shift: 0.18 (normal), 0.56 (fullPath). Same map, extra `probe_passFullPath==1`: largest fitted $|\Delta|$ 0.114 (normal) vs 0.0083 (fullPath).

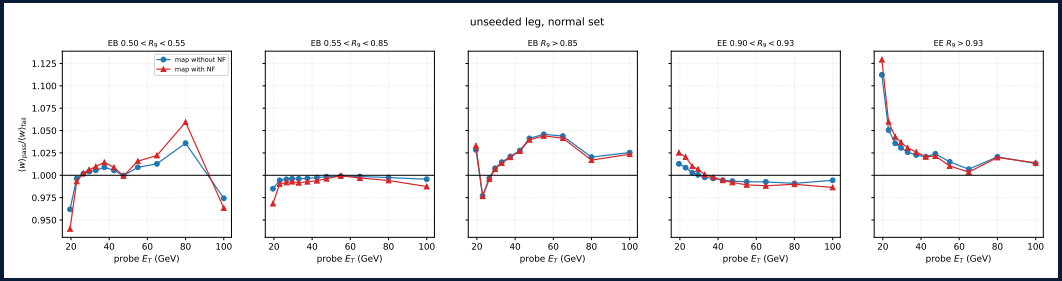
The six largest fitted shifts, cell by cell



First-order contribution of each cell to $\Delta\varepsilon_{\mathrm{count}}$; Σ cells vs exact counting shift in each title.

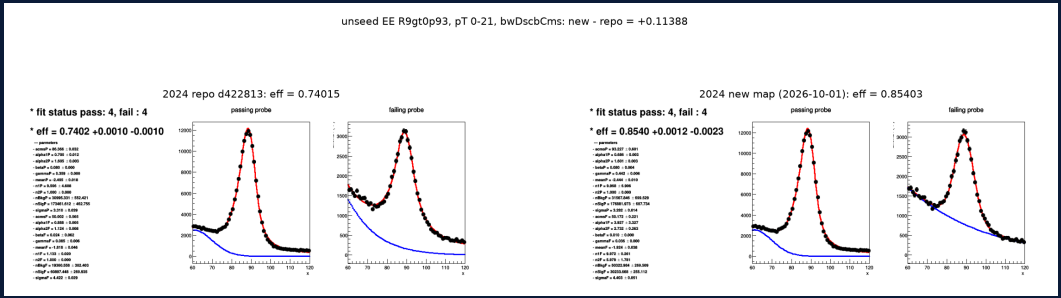
In 4 of the 6, $\Delta\varepsilon_{\mathrm{count}}$ has the opposite sign to the fitted shift; in all 6, $|\Delta\varepsilon_{\mathrm{count}}| \leq 0.0048$.

Mean weight of passing vs failing probes — unseeded



$\varepsilon_{\mathrm{count}}$ changes only through $\langle w \rangle_{\mathrm{pass}} / \langle w \rangle_{\mathrm{fail}}$; shown for both maps.

The largest shift in the fit: unseeded EE $R_9 > 0.93$, $E_T < 21$ GeV

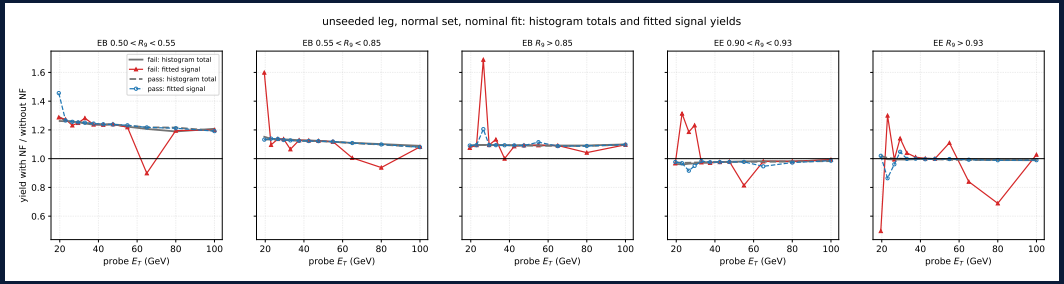


Left without NF, right with NF. Both print `fit status pass: 4, fail: 4`. Failing-probe fit: background yield 19 361 → 50 323, signal yield 60 897 → 30 234 (values printed on the plots). Counting shift for this bin: +0.0030; fitted shift +0.114.

From map to fitted efficiency: three stages

- **1. Map → weighted yields.** Per fitter bin, pass and fail: total of the mass histogram in 60–120 GeV, with NF / without NF. Equal to the yields rebuilt from per-cell probe counts to 1.5×10^{-5} (relative).
- **2. Weighted yields → mass templates.** The pass / fail histograms the fitter fitted, without vs with NF: shape compared by the total-variation distance $\mathrm{TV} = \frac{1}{2} \sum |p_{\mathrm{with\,NF}} - p_{\mathrm{without\,NF}}|$ (unit-area histograms); sideband-subtracted efficiency $\varepsilon_{\mathrm{sb}}$ (window 80–100 GeV minus sidebands 60–70 and 110–120 GeV), no fit.
- **3. Mass templates → fitted yields.** N_{sig}, N_{bkg} and stored fit status of every pass and fail fit, read from the fit-result files. $N_{\mathrm{sig}}^{\mathrm{pass}} / (N_{\mathrm{sig}}^{\mathrm{pass}} + N_{\mathrm{sig}}^{\mathrm{fail}})$ reproduces the published efficiency of both sets to 5×10^{-6}.
- $\varepsilon_{\mathrm{sb}}$ is not validated as an efficiency; only its change between the two maps is used.

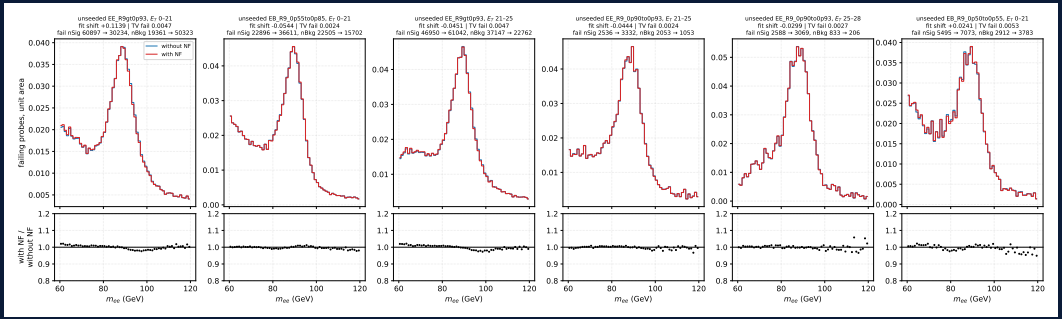
Stage 1 and 3: histogram totals vs fitted signal yields — unseeded



Yield with NF / without NF. Grey: histogram total in 60–120 GeV (fail solid, pass dashed). Red: fitted fail signal. Blue: fitted pass signal. Normal set, nominal fit.

Normal set, largest change: histogram total 29% (fail), 26% (pass); fitted signal 69% (fail), 46% (pass).

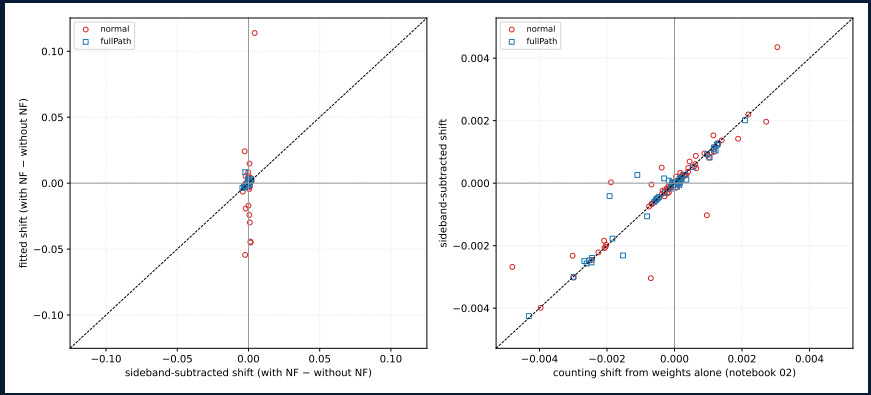
Stage 2: fail mass templates of the six largest fitted shifts



Failing-probe mass histograms, 60–120 GeV, unit area. Titles: fitted shift, TV distance, fitted fail N_{sig} and N_{bkg} without NF → with NF.

Largest TV distance over all bins: fail 0.043 (normal), 0.059 (fullPath); pass 0.014, 0.014.

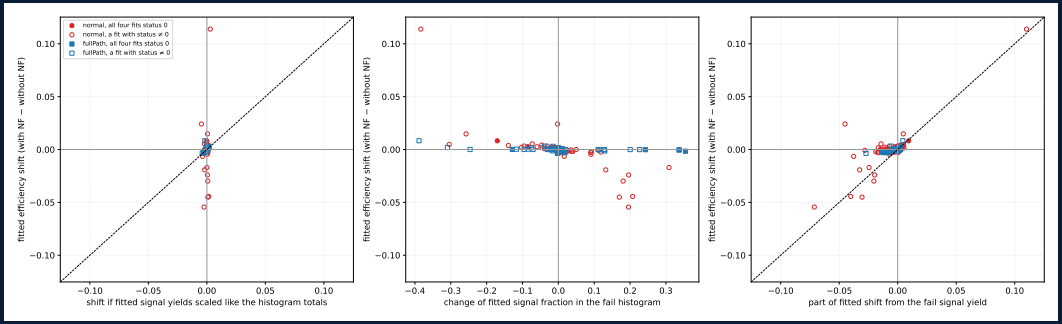
Stage 2: sideband-subtracted efficiency, no fit



Left: fitted shift vs sideband-subtracted shift. Right: sideband-subtracted shift vs counting shift from the weights alone.

Largest $|\Delta\varepsilon_{\mathrm{sb}}|$: 0.0044 (normal), 0.0043 (fullPath). Correlation with the counting shift: 0.91, 0.97; with the fitted shift: 0.28, 0.50.

Stage 3: fitted shift vs the fitted yields



Left: shift if the fitted signal yields had scaled like the histogram totals. Middle: change of the fitted signal fraction of the fail histogram. Right: part of the fitted shift from the fail signal yield. Open markers: at least one of the four fits (pass/fail, without/with NF) stores status $\neq 0$.

Correlation of the fitted shift with: yields scaled like totals 0.18 (normal), 0.55 (fullPath); fail-signal part 0.83, 0.65.

Stage 3: stored fit status

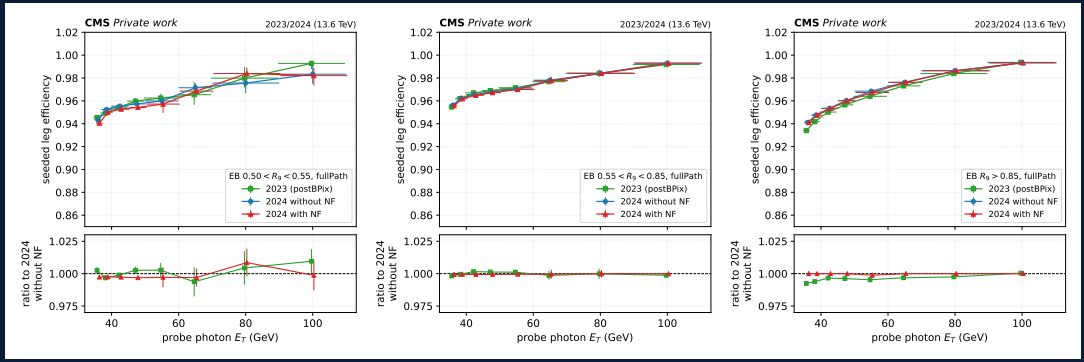
Fits with stored status $\neq 0$	RooCMSShape (nominal) pass	RooCMSShape (nominal) fail	Exponential pass	Exponential fail
2024 without NF, normal	49 / 105	52 / 105	43 / 105	45 / 105
2024 with NF, normal	49 / 105	50 / 105	40 / 105	48 / 105
2024 without NF, fullPath	49 / 105	28 / 105	41 / 105	19 / 105
2024 with NF, fullPath	48 / 105	23 / 105	44 / 105	21 / 105

- Stored status values: 0 and 4 only. Stored covariance quality: -1 in all 1680 fits.
- Nominal fit, normal set: 20 of 105 bins have status 0 in all four fits (pass/fail, without/with NF); largest fitted $|\Delta|$ there 0.0082, in the other 85 bins 0.114.
- fullPath: 29 of 105 bins; largest $|\Delta|$ 0.0030 vs 0.0083.
- All 10 normal-set bins with fitted $|\Delta| > 0.01$ contain a fit with stored status 4; no bin has $|\Delta\varepsilon_{\mathrm{sb}}| > 0.01$.

Backup

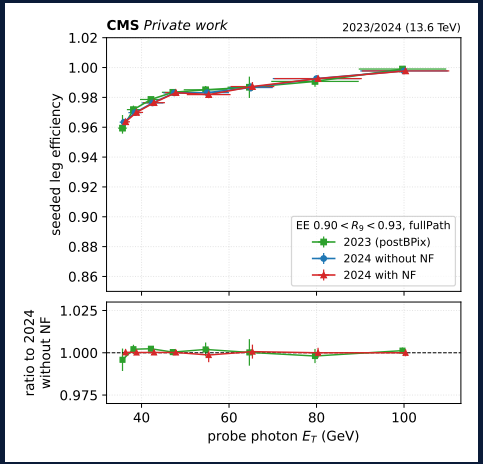
fullPath and extra views

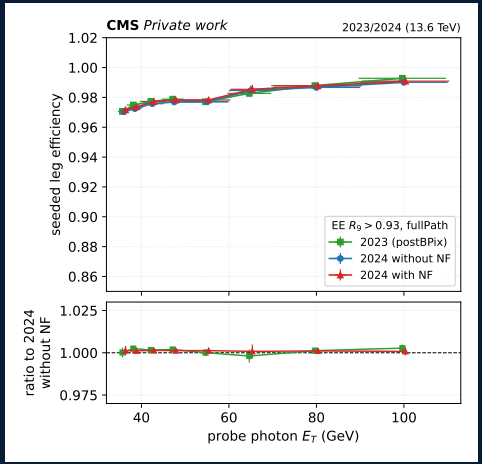
Backup: fullPath — seeded leg, EB bins



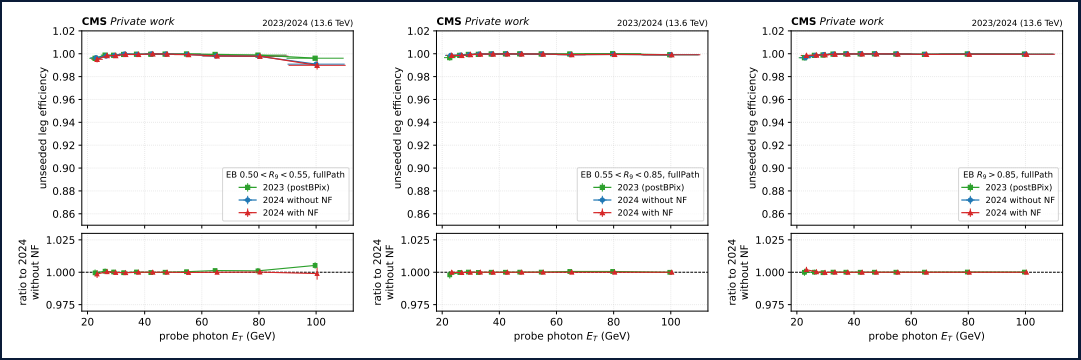
2023 fullPath CSV, 2024 without NF `etaR9rw_withPresel_fullPath`, 2024 with NF `HLT_2024_withPresel_fullPath_SS`.

Backup: fullPath — seeded leg, EE bins

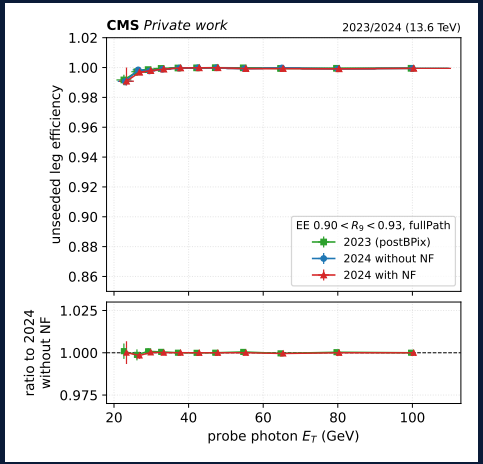


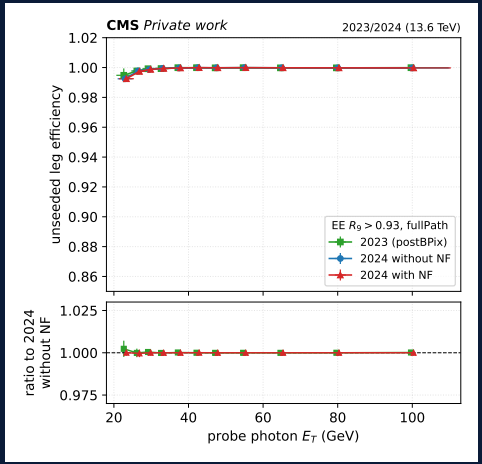


Backup: fullPath — unseeded leg, EB bins

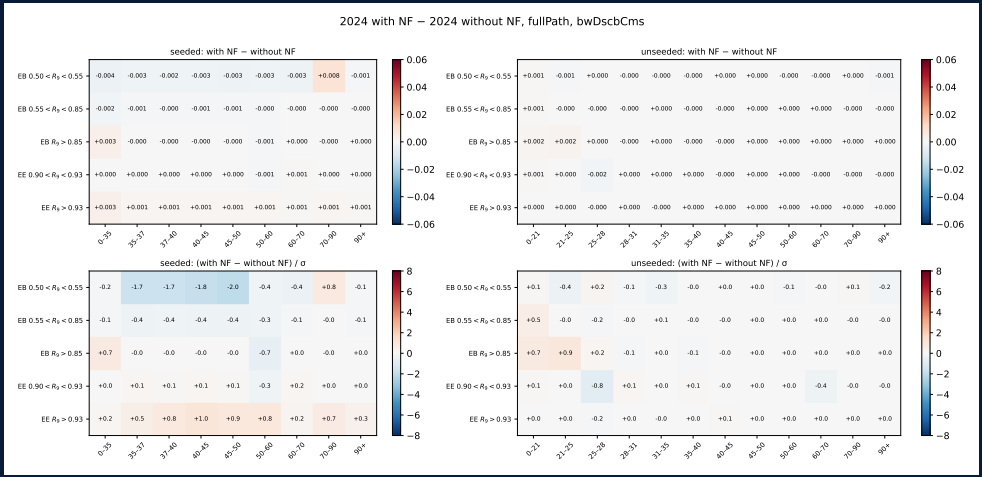


Backup: fullPath — unseeded leg, EE bins



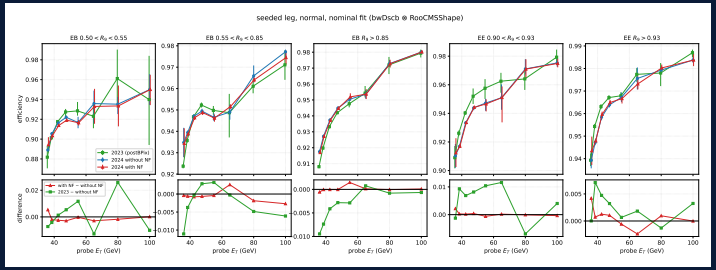


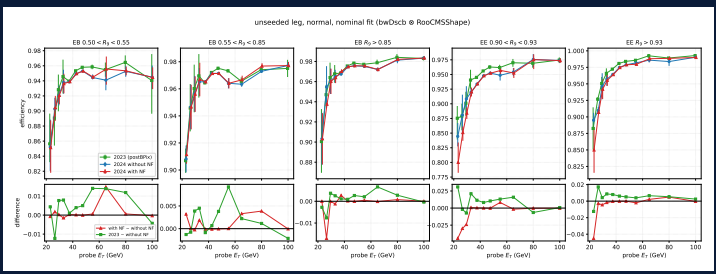
Backup: fullPath — with NF − without NF, every bin



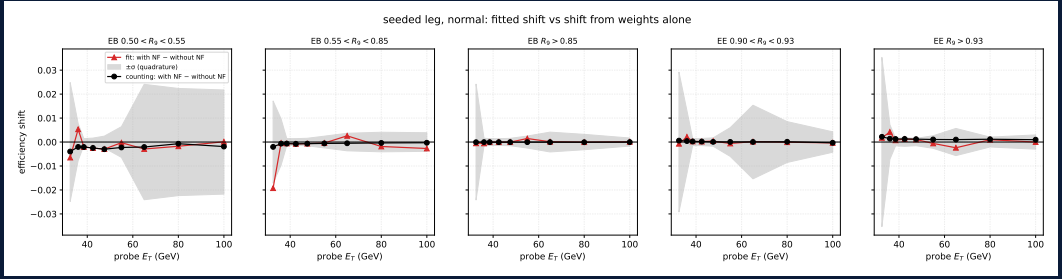
Same layout as the normal-set heatmap.

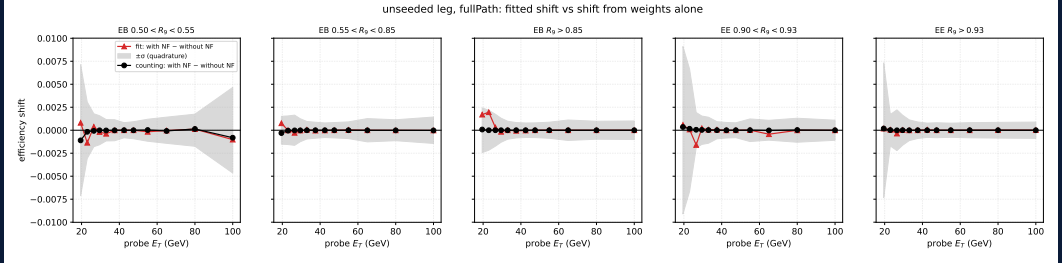
Backup: one-page overviews, normal set



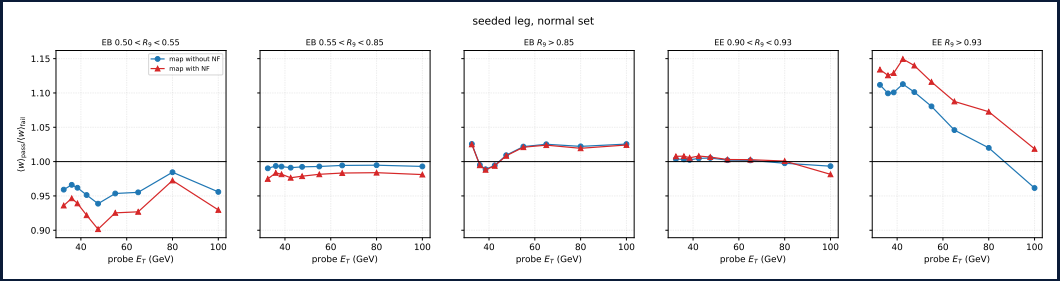


Backup: fitted vs counting shift — seeded, and fullPath

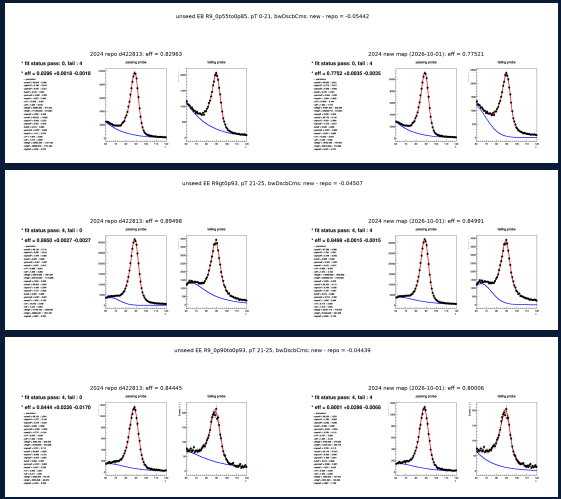




Backup: mean weight of passing vs failing probes — seeded

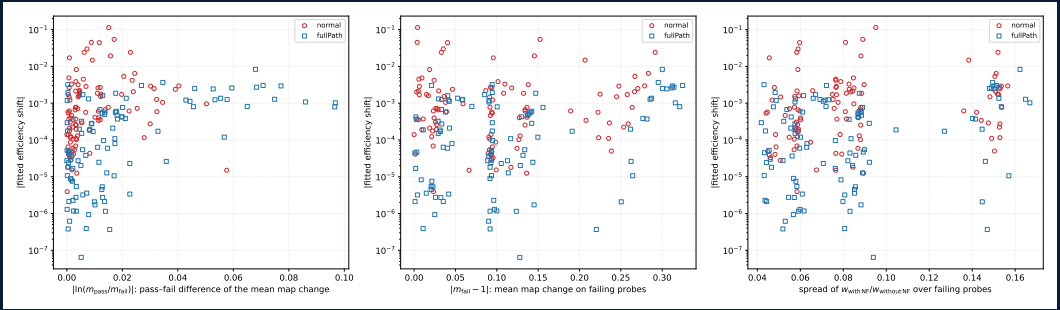


Backup: fit pairs for shifts 2–4



Left without NF, right with NF. Unseeded EB $0.55 < R_9 < 0.85$ 0–21; EE $R_9 > 0.93$ 21–25; EE $0.90 < R_9 < 0.93$ 21–25 (nominal fit).

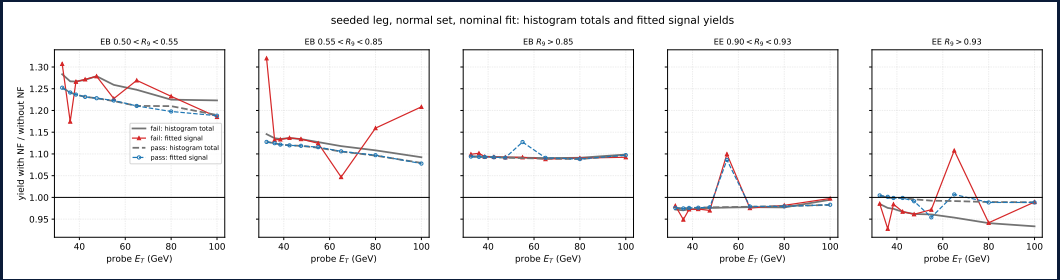
Backup: size of the map change vs size of the fitted shift



Per fitter bin. m_{pass}, m_{fail}: weighted yield with NF / without NF. Right: spread of the per-cell weight ratio over the failing probes.

Spearman correlation with |fitted shift|, normal set: 0.46, 0.03, 0.26 (left to right); fullPath: 0.53, 0.32, 0.24.

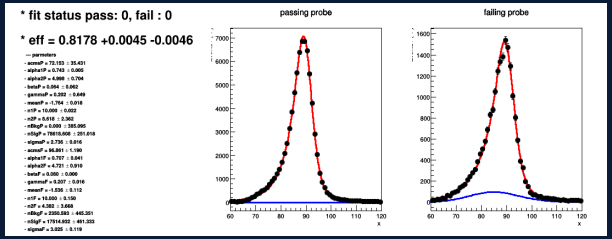
Backup: histogram totals vs fitted signal yields — seeded



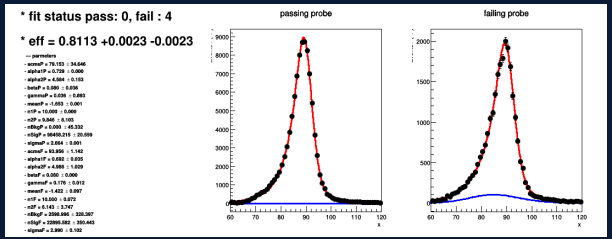
Same layout as the unseeded slide.

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $0 < E_T < 35$ GeV

without NF



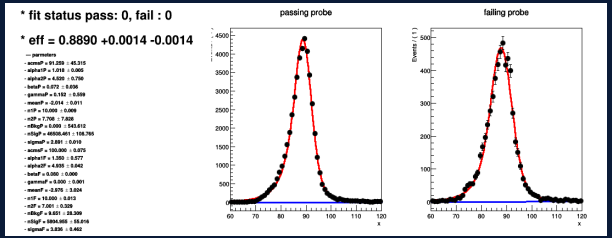
with NF



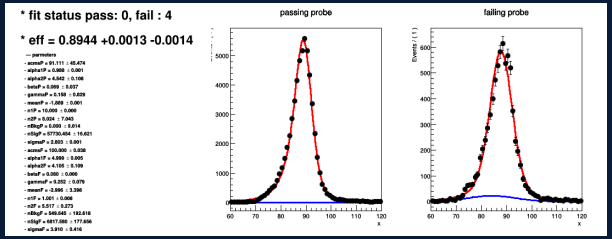
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $35 < E_T < 37$ GeV

without NF



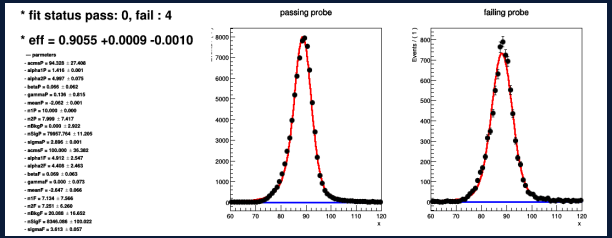
with NF



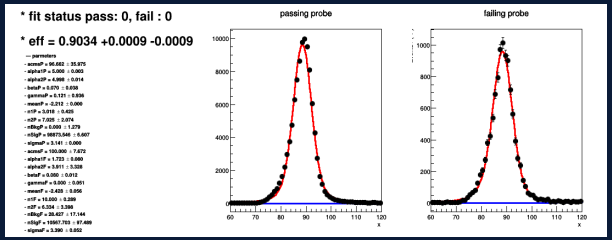
efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $37 < E_T < 40$ GeV

without NF



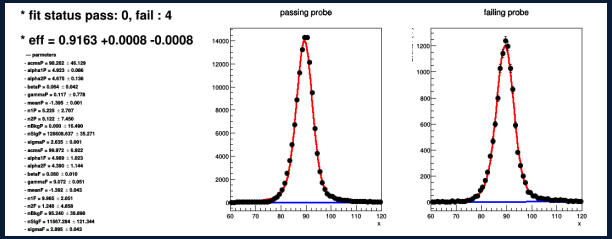
with NF



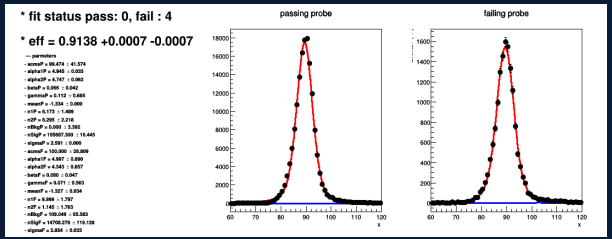
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $40 < E_T < 45$ GeV

without NF



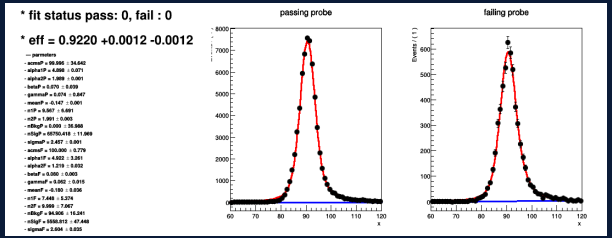
with NF



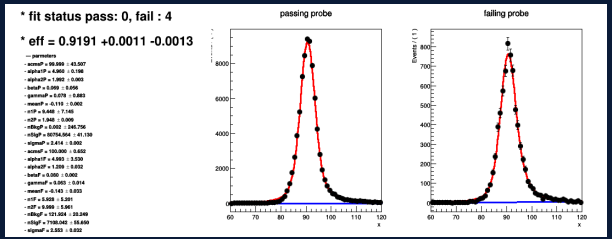
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $45 < E_T < 50$ GeV

without NF



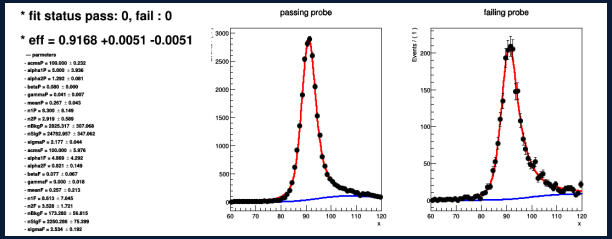
with NF



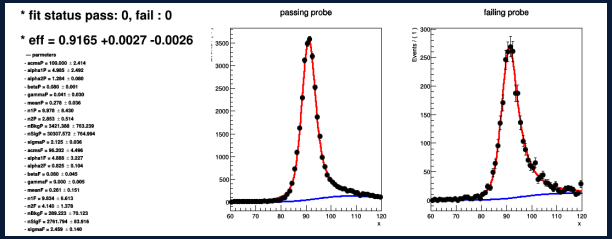
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $50 < E_T < 60$ GeV

without NF



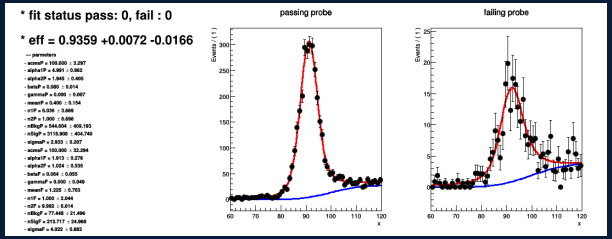
with NF



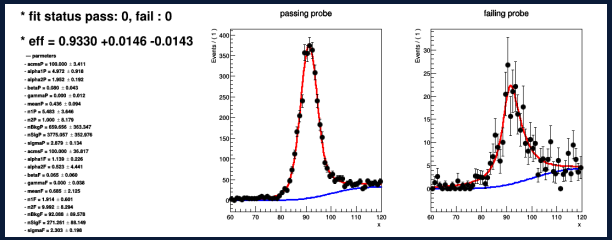
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $60 < E_T < 70$ GeV

without NF



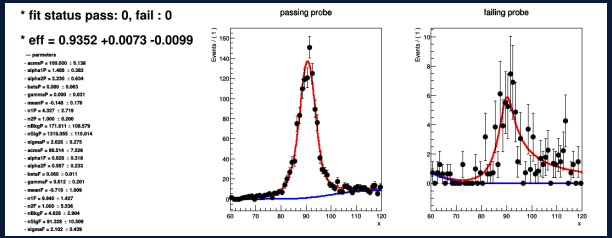
with NF



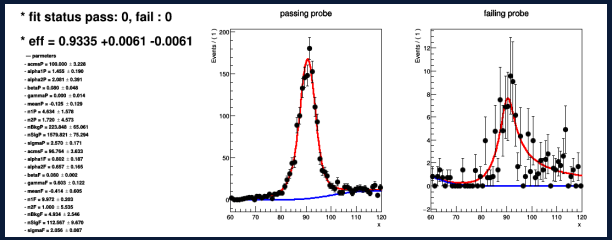
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $70 < E_T < 90$ GeV

without NF



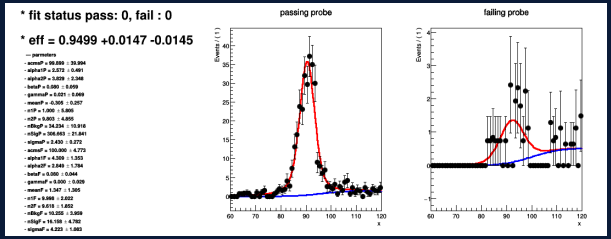
with NF



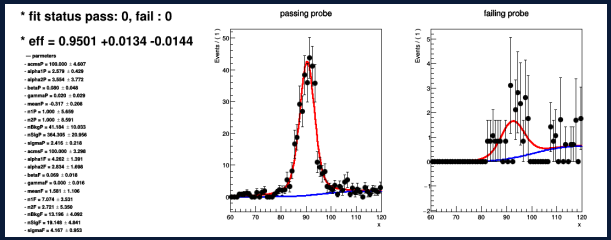
◂ efficiency

Fits: seeded leg, EB $0.50 < R_9 < 0.55$, $E_T > 90$ GeV

without NF



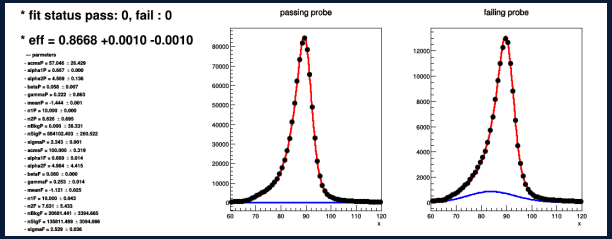
with NF



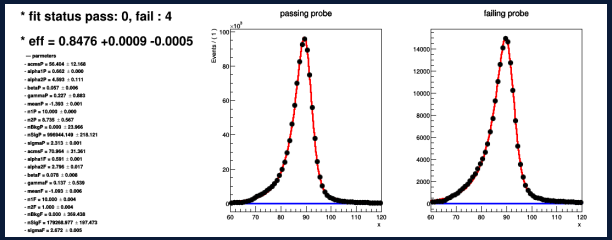
◀ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $0 < E_T < 35$ GeV

without NF



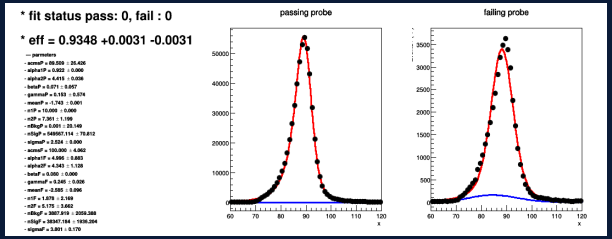
with NF



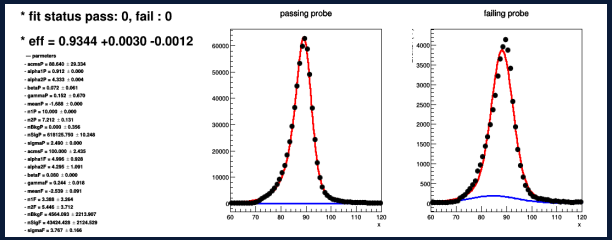
◂ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $35 < E_T < 37$ GeV

without NF



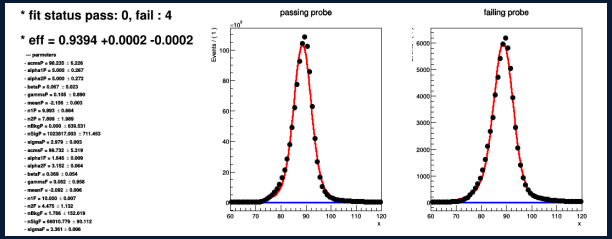
with NF



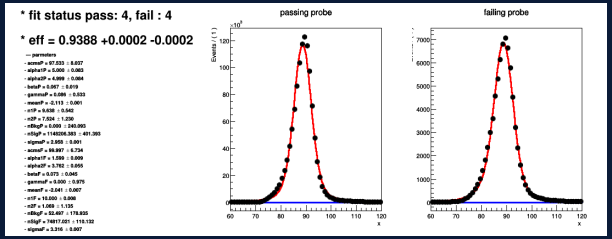
◂ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $37 < E_T < 40$ GeV

without NF



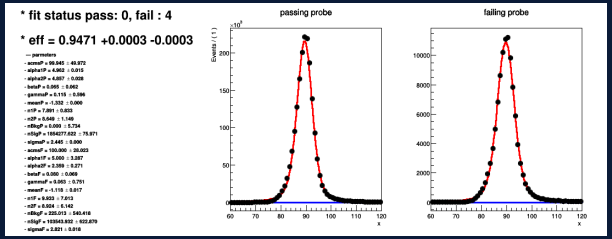
with NF



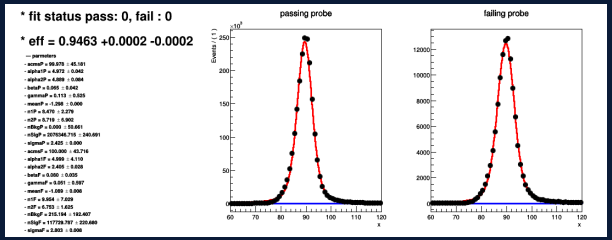
◂ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $40 < E_T < 45$ GeV

without NF



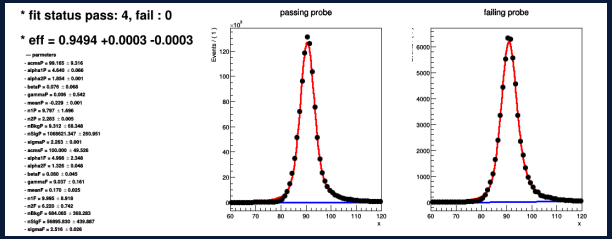
with NF



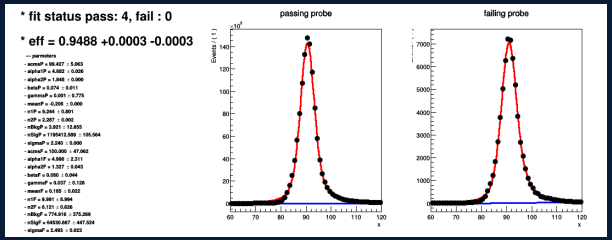
◂ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $45 < E_T < 50$ GeV

without NF



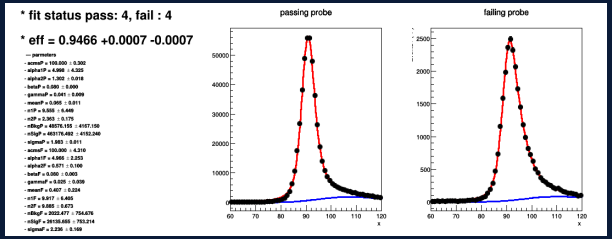
with NF



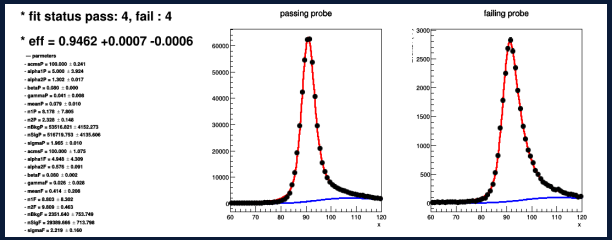
◀ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $50 < E_T < 60$ GeV

without NF



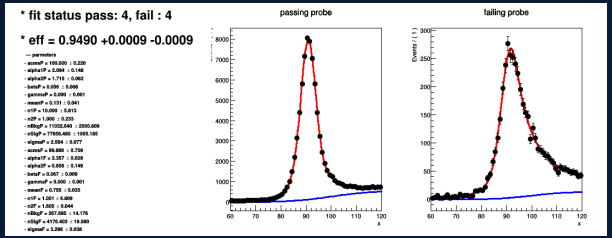
with NF



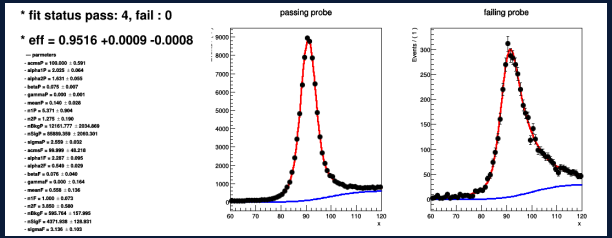
◀ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $60 < E_T < 70$ GeV

without NF



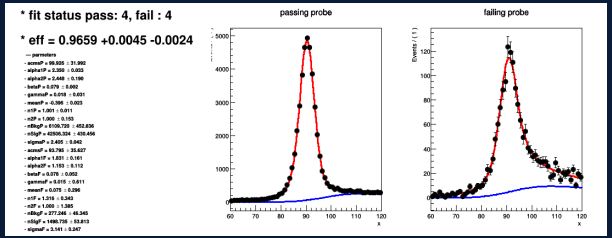
with NF



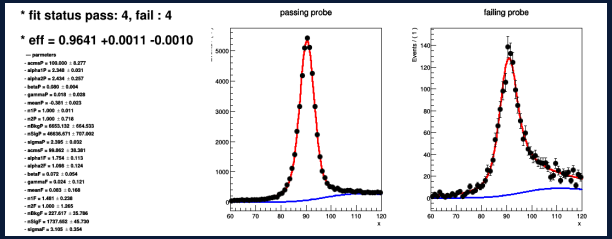
◂ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $70 < E_T < 90$ GeV

without NF



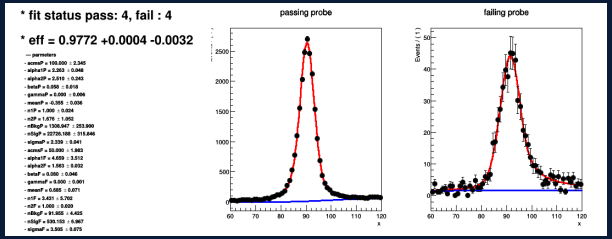
with NF



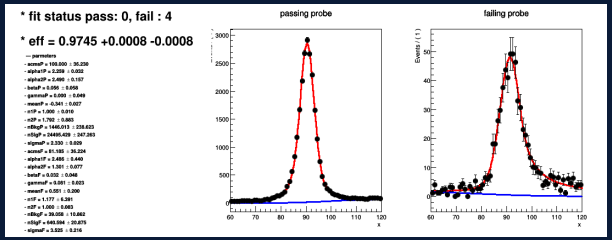
◂ efficiency

Fits: seeded leg, EB $0.55 < R_9 < 0.85$, $E_T > 90$ GeV

without NF



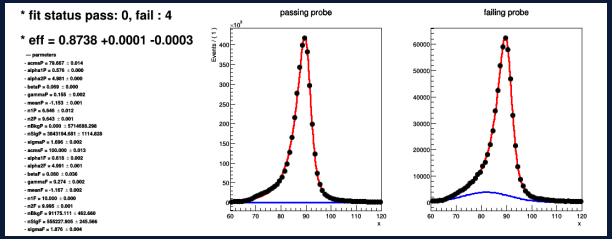
with NF



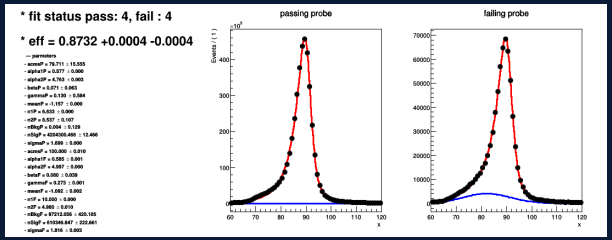
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $0 < E_T < 35$ GeV

without NF



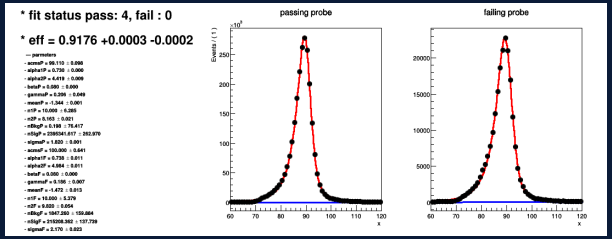
with NF



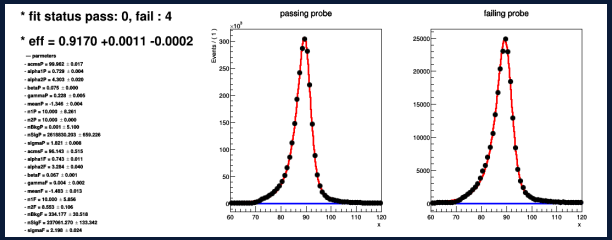
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $35 < E_T < 37$ GeV

without NF



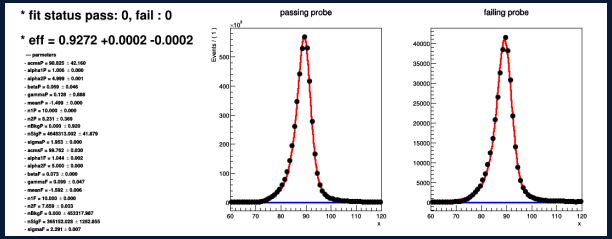
with NF



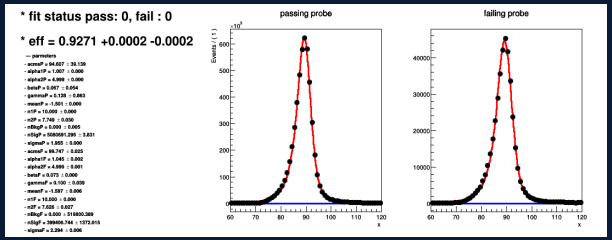
◀ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $37 < E_T < 40$ GeV

without NF



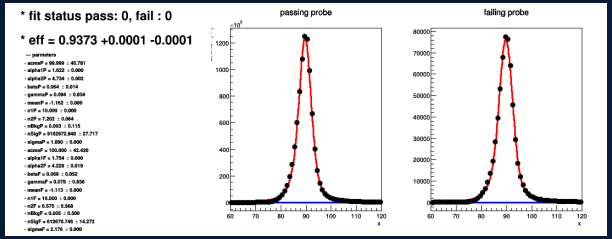
with NF



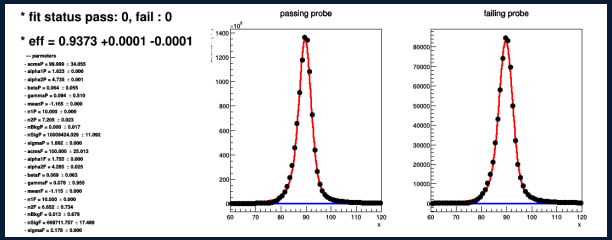
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $40 < E_T < 45$ GeV

without NF



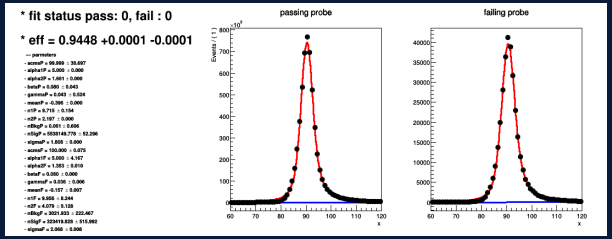
with NF



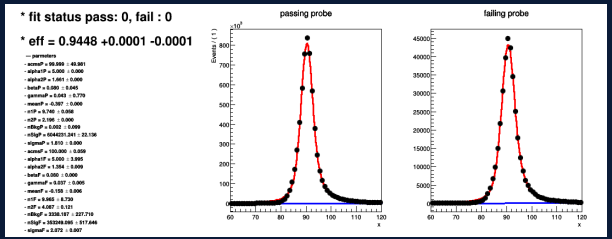
efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $45 < E_T < 50$ GeV

without NF



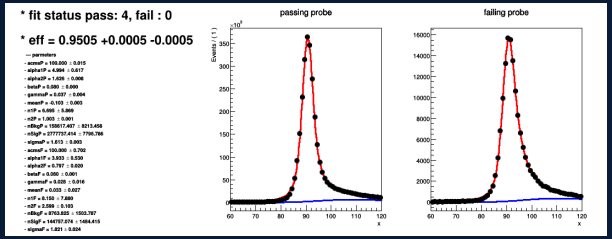
with NF



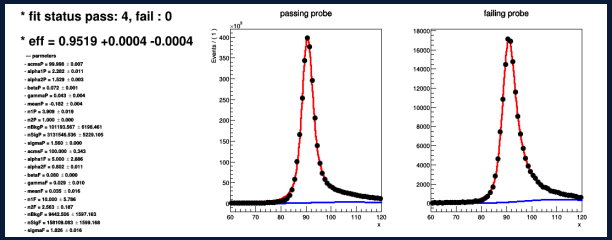
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $50 < E_T < 60$ GeV

without NF



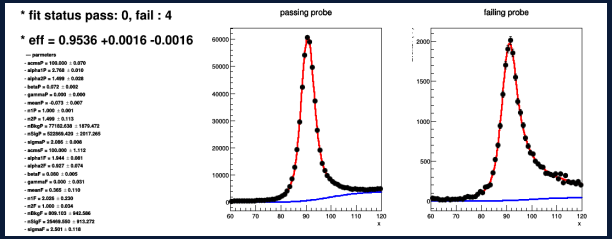
with NF



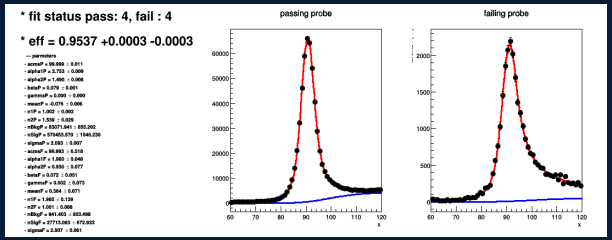
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $60 < E_T < 70$ GeV

without NF



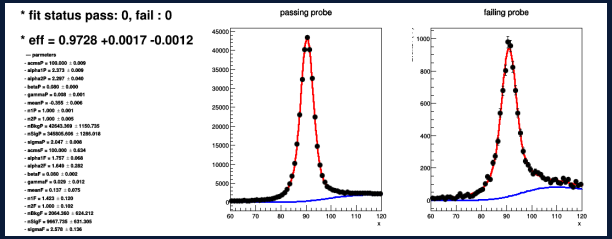
with NF



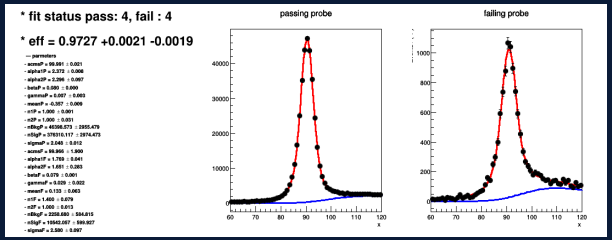
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $70 < E_T < 90$ GeV

without NF



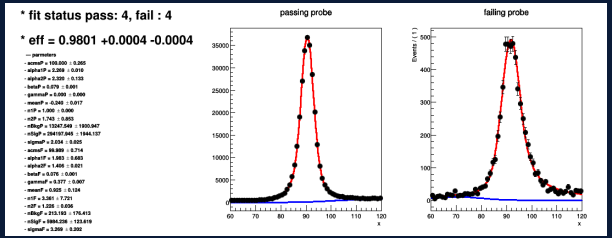
with NF



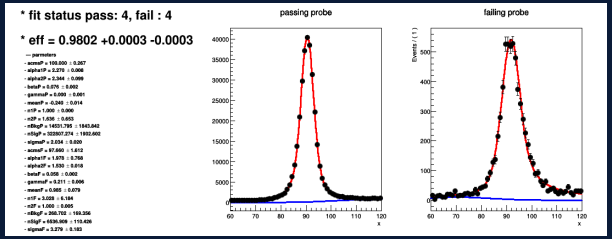
◂ efficiency

Fits: seeded leg, EB $R_9 > 0.85$, $E_T > 90$ GeV

without NF



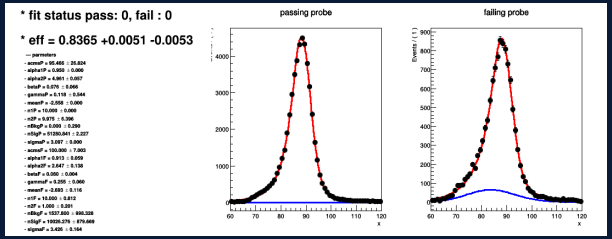
with NF



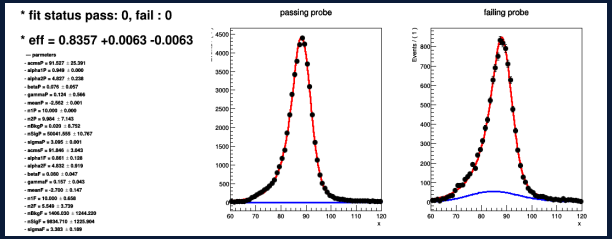
◂ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $0 < E_T < 35$ GeV

without NF



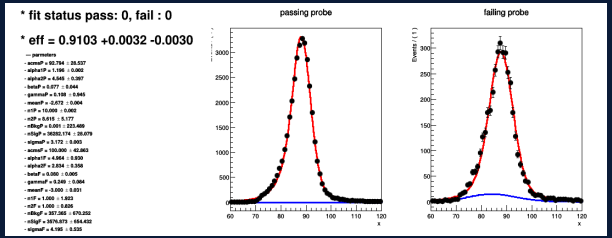
with NF



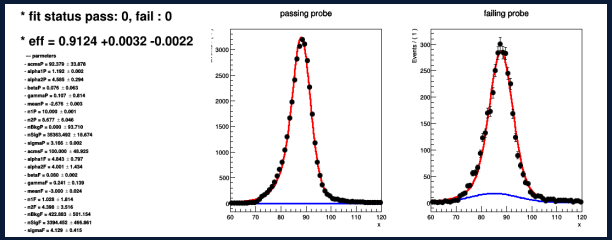
◂ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $35 < E_T < 37$ GeV

without NF



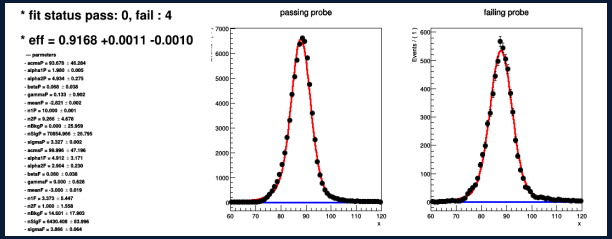
with NF



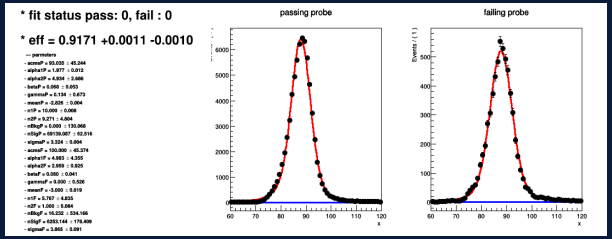
◂ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $37 < E_T < 40$ GeV

without NF



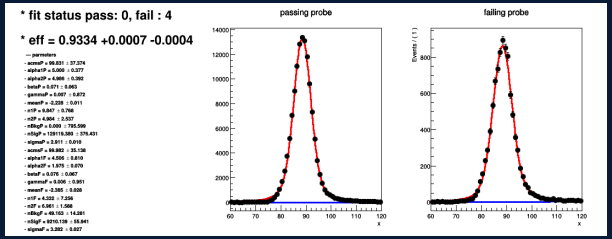
with NF



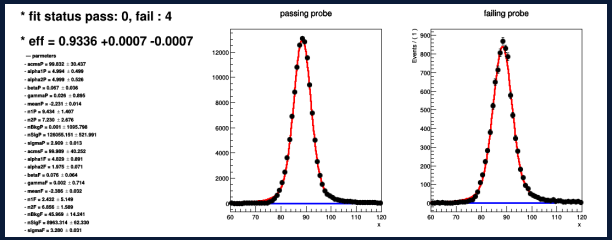
◂ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $40 < E_T < 45$ GeV

without NF



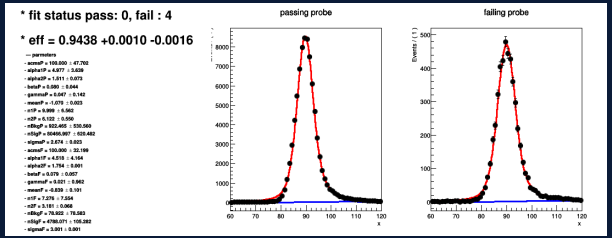
with NF



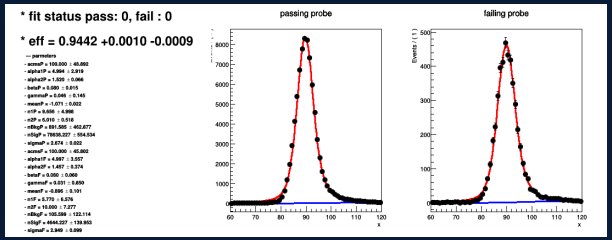
◀ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $45 < E_T < 50$ GeV

without NF



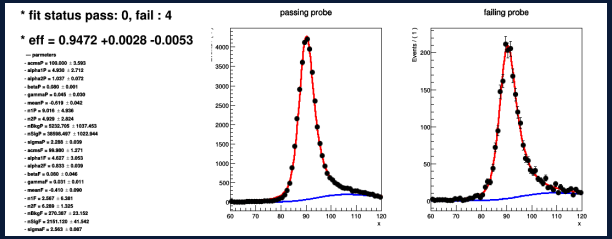
with NF



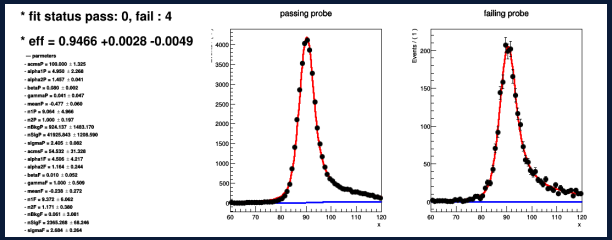
◂ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $50 < E_T < 60$ GeV

without NF



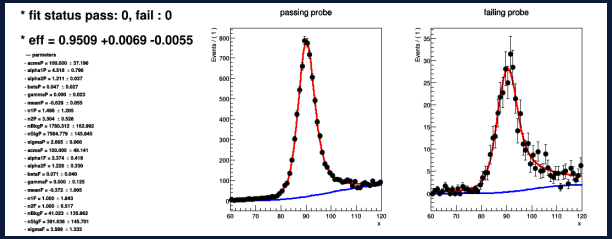
with NF



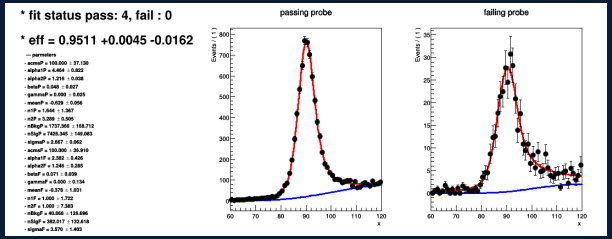
◀ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $60 < E_T < 70$ GeV

without NF



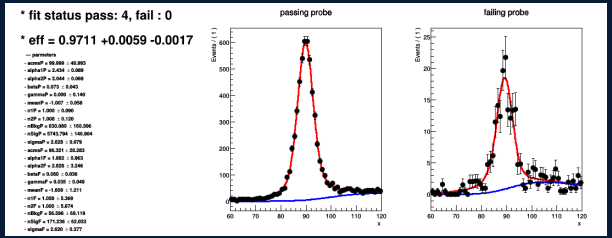
with NF



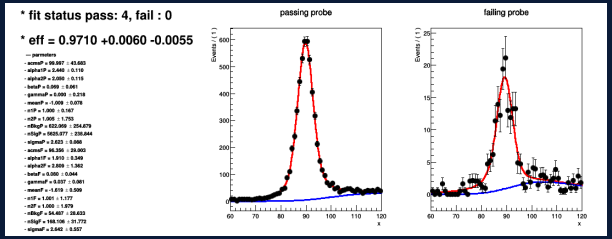
◂ efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $70 < E_T < 90$ GeV

without NF



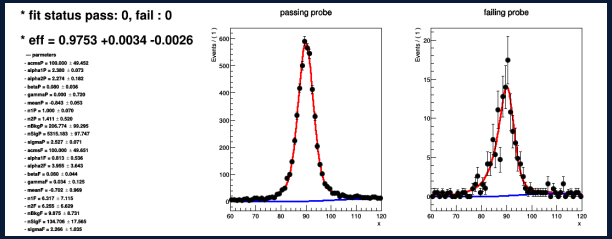
with NF



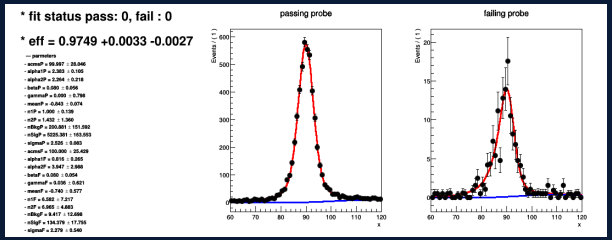
efficiency

Fits: seeded leg, EE $0.90 < R_9 < 0.93$, $E_T > 90$ GeV

without NF



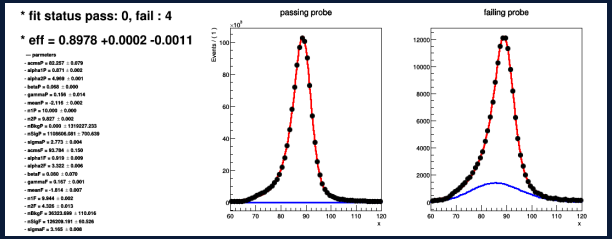
with NF



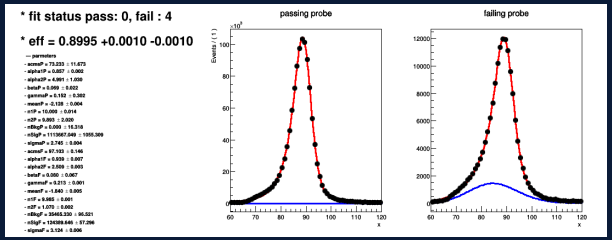
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $0 < E_T < 35$ GeV

without NF



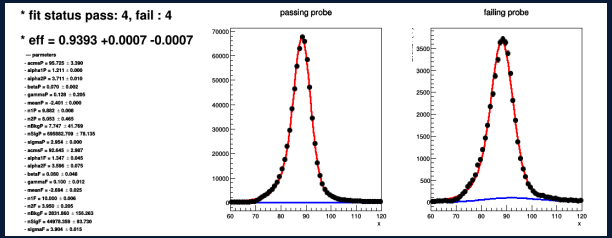
with NF



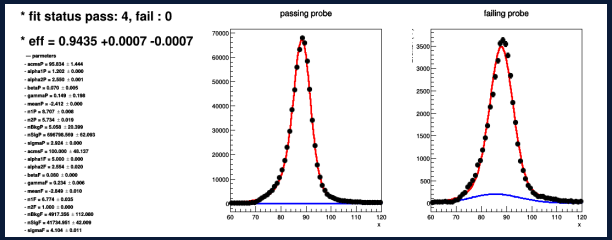
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $35 < E_T < 37$ GeV

without NF



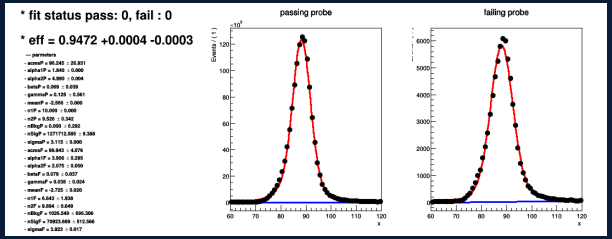
with NF



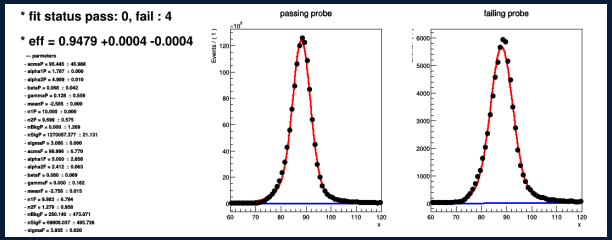
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $37 < E_T < 40$ GeV

without NF



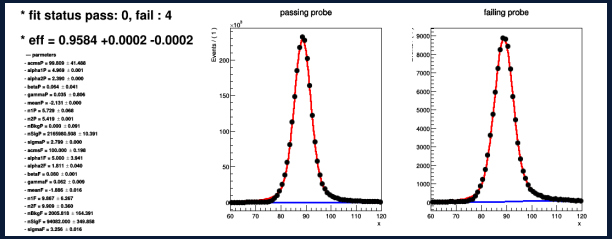
with NF



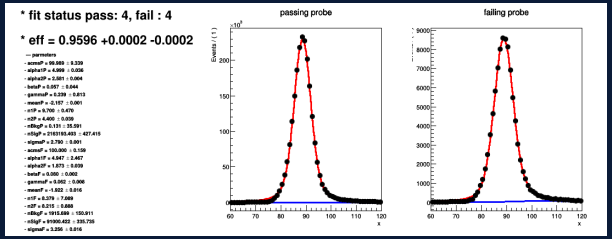
efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $40 < E_T < 45$ GeV

without NF



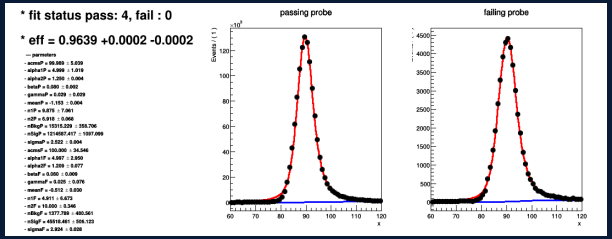
with NF



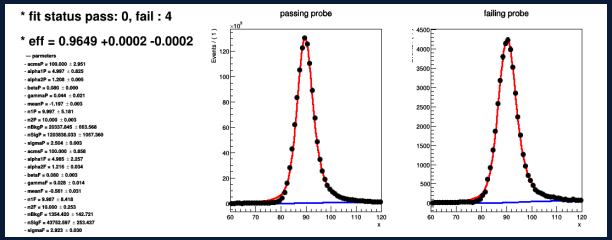
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $45 < E_T < 50$ GeV

without NF



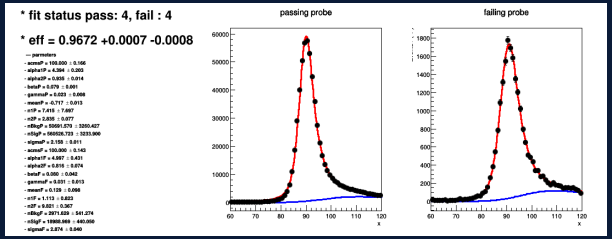
with NF



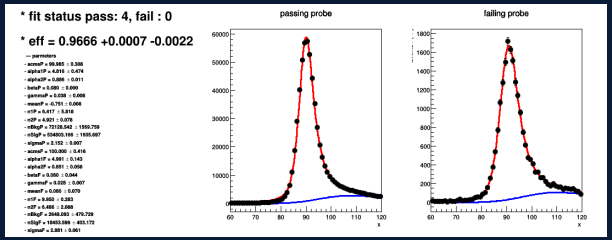
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $50 < E_T < 60$ GeV

without NF



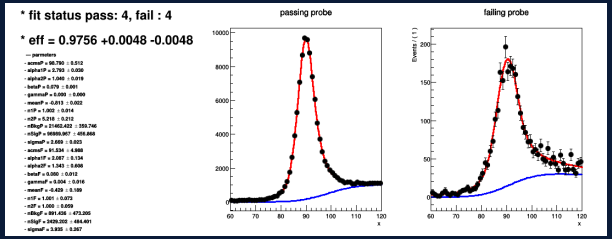
with NF



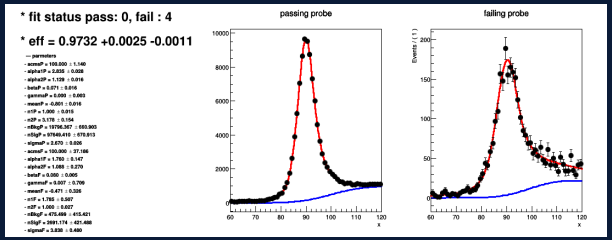
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $60 < E_T < 70$ GeV

without NF



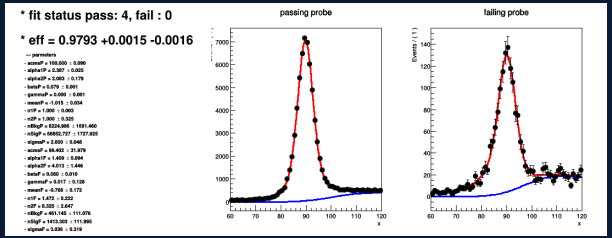
with NF



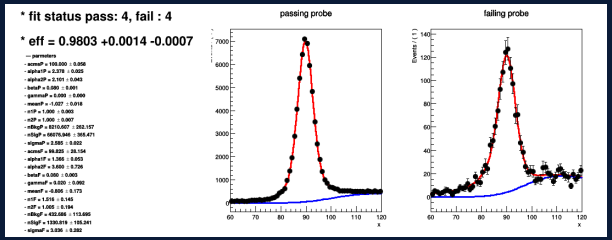
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $70 < E_T < 90$ GeV

without NF



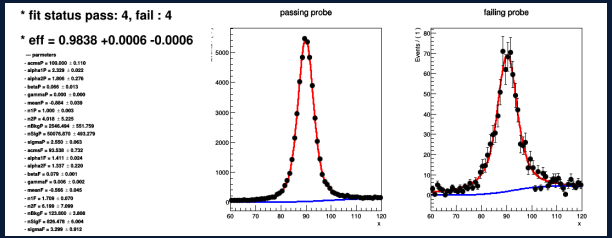
with NF



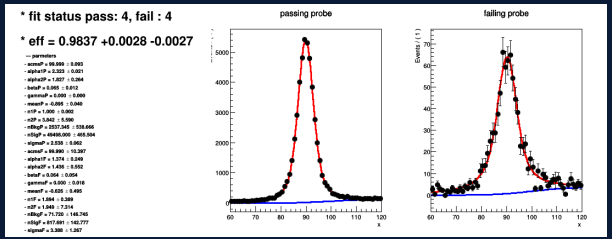
◂ efficiency

Fits: seeded leg, EE $R_9 > 0.93$, $E_T > 90$ GeV

without NF



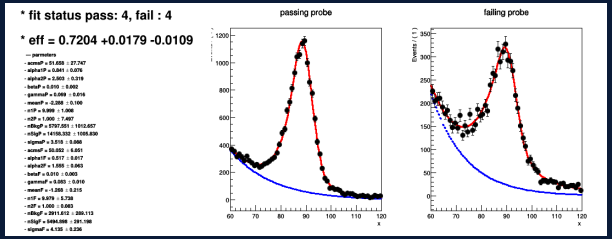
with NF



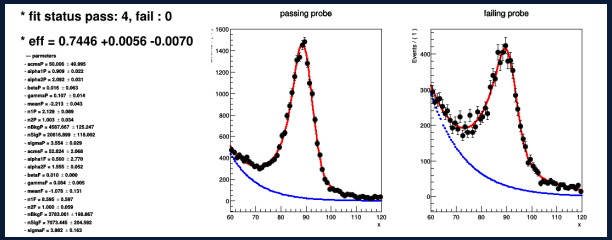
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $0 < E_T < 21$ GeV

without NF



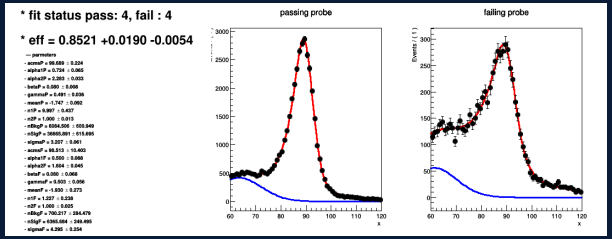
with NF



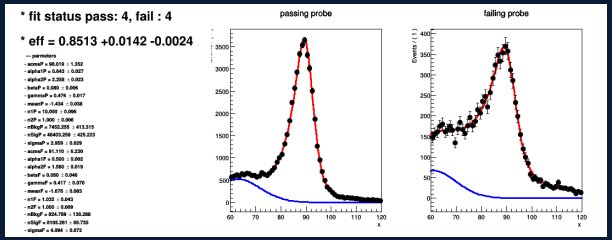
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $21 < E_T < 25$ GeV

without NF



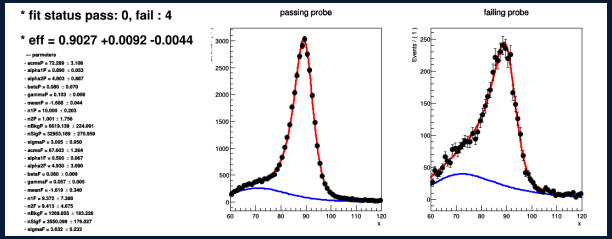
with NF



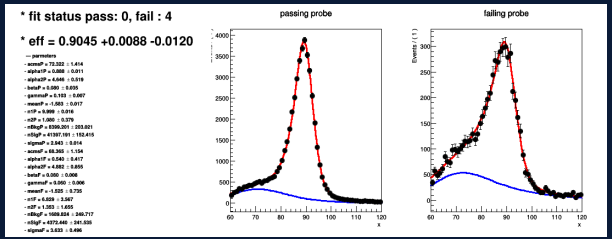
◀ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $25 < E_T < 28$ GeV

without NF



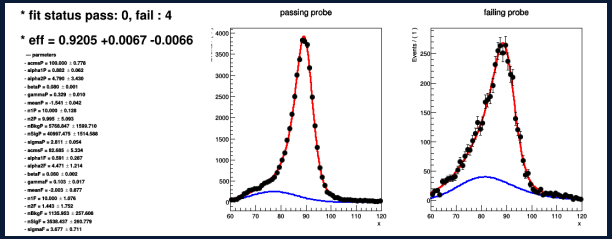
with NF



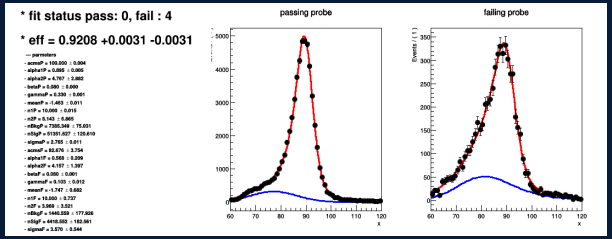
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $28 < E_T < 31$ GeV

without NF



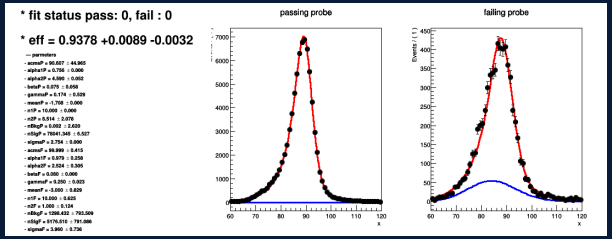
with NF



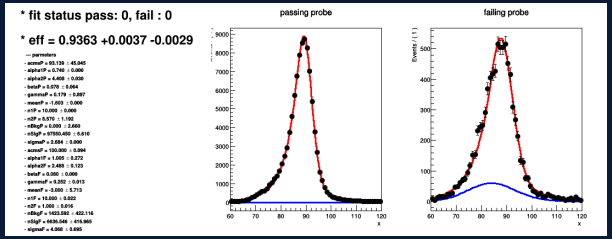
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $31 < E_T < 35$ GeV

without NF



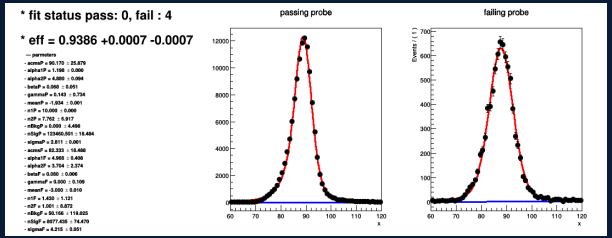
with NF



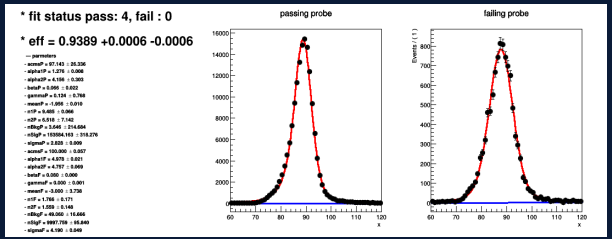
efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $35 < E_T < 40$ GeV

without NF



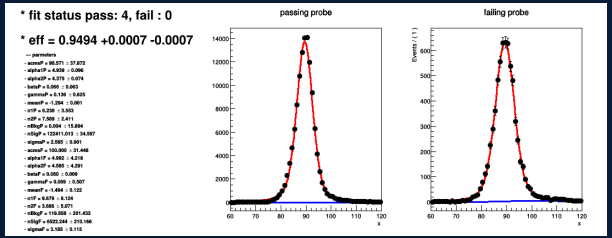
with NF



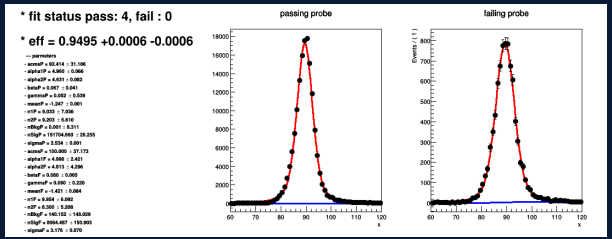
◀ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $40 < E_T < 45$ GeV

without NF



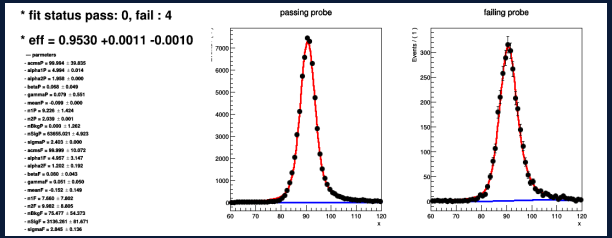
with NF



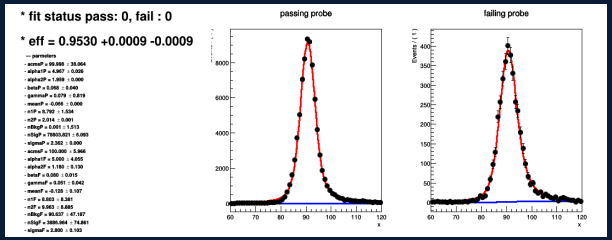
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $45 < E_T < 50$ GeV

without NF



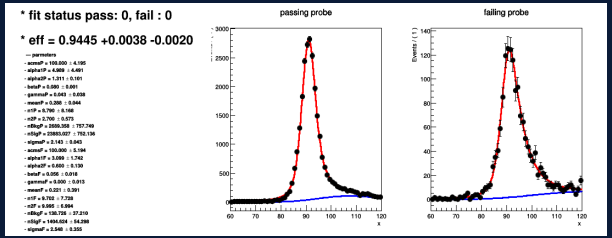
with NF



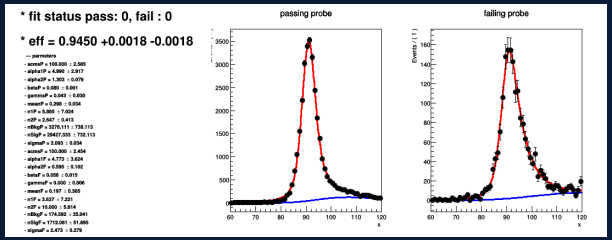
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $50 < E_T < 60$ GeV

without NF



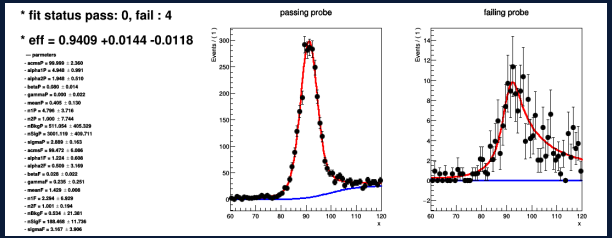
with NF



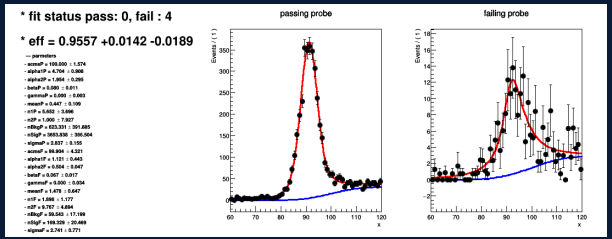
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $60 < E_T < 70$ GeV

without NF



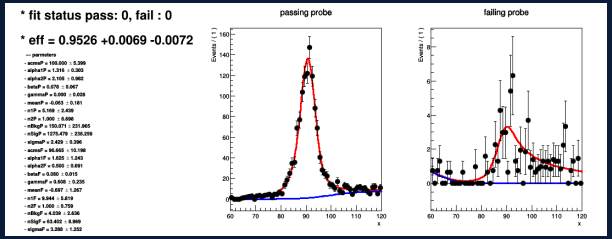
with NF



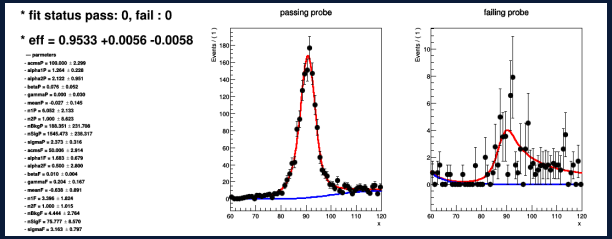
◂ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $70 < E_T < 90$ GeV

without NF



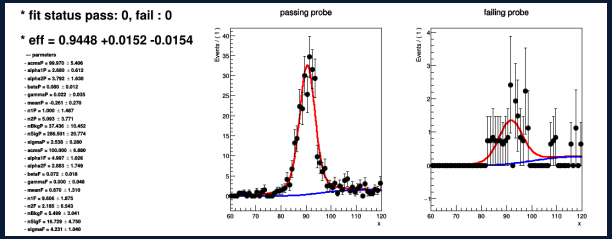
with NF



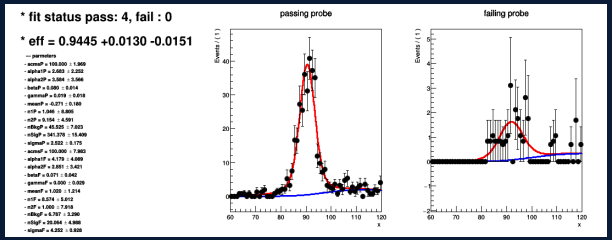
◀ efficiency

Fits: unseeded leg, EB $0.50 < R_9 < 0.55$, $E_T > 90$ GeV

without NF



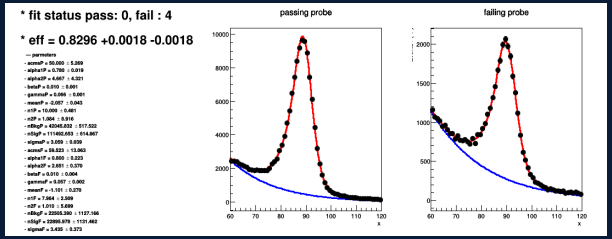
with NF



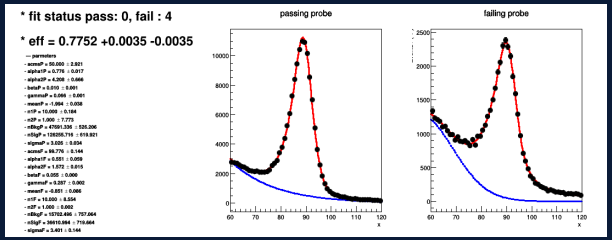
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $0 < E_T < 21$ GeV

without NF



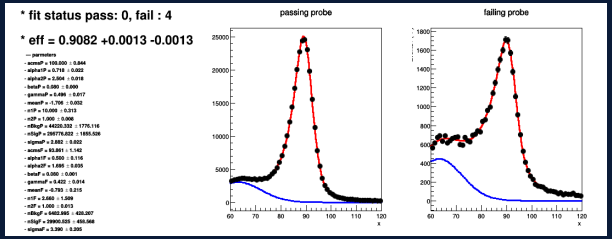
with NF



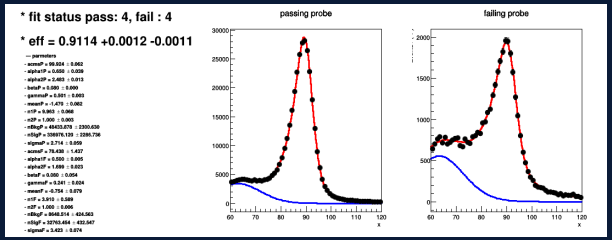
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $21 < E_T < 25$ GeV

without NF



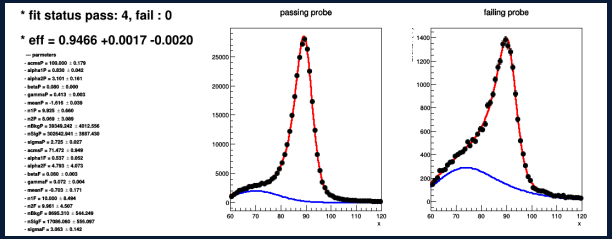
with NF



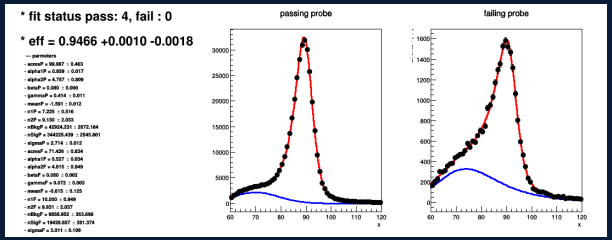
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $25 < E_T < 28$ GeV

without NF



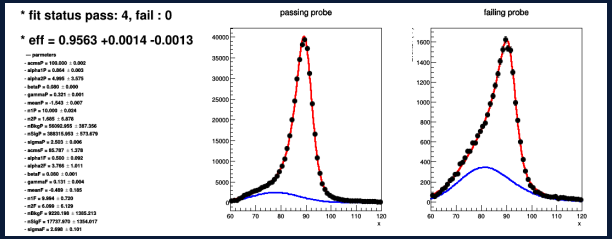
with NF



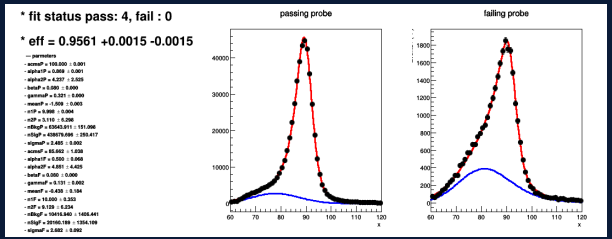
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $28 < E_T < 31$ GeV

without NF



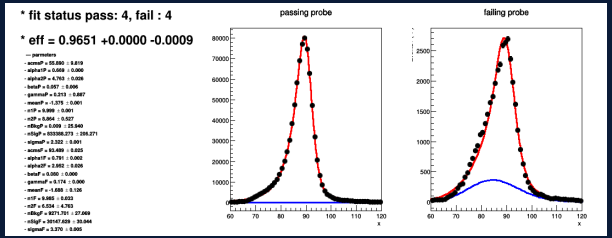
with NF



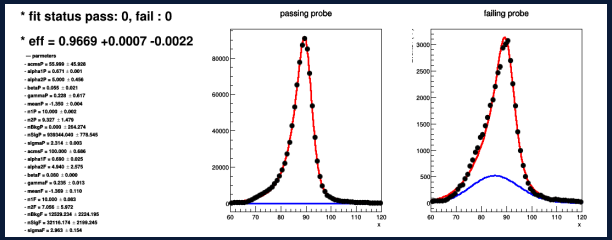
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $31 < E_T < 35$ GeV

without NF



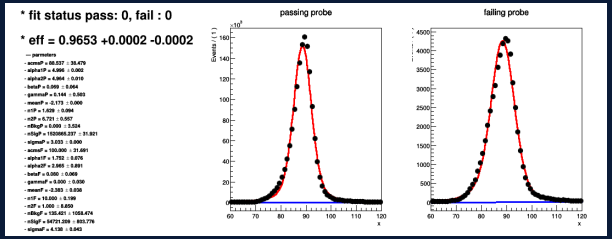
with NF



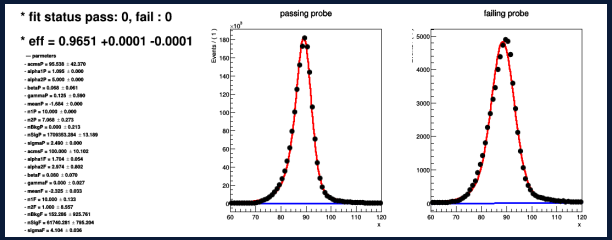
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $35 < E_T < 40$ GeV

without NF



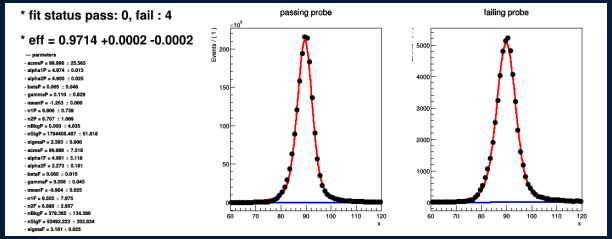
with NF



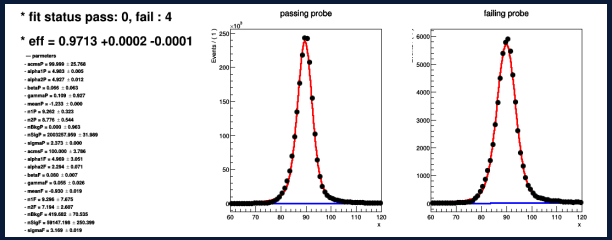
◀ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $40 < E_T < 45$ GeV

without NF



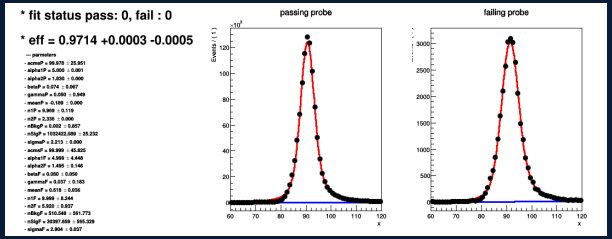
with NF



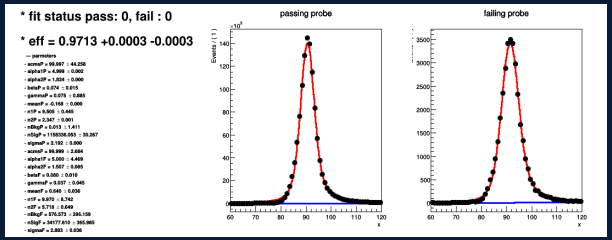
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $45 < E_T < 50$ GeV

without NF



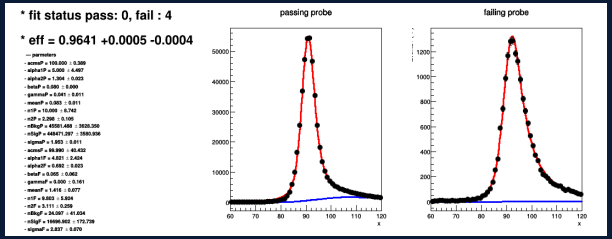
with NF



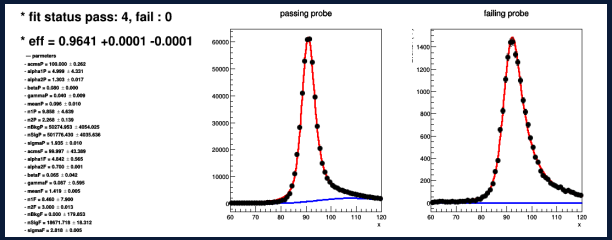
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $50 < E_T < 60$ GeV

without NF



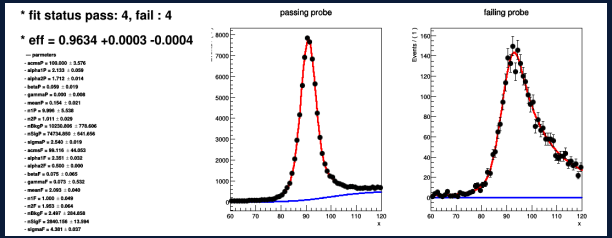
with NF



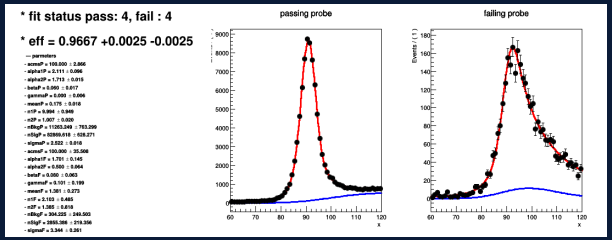
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $60 < E_T < 70$ GeV

without NF



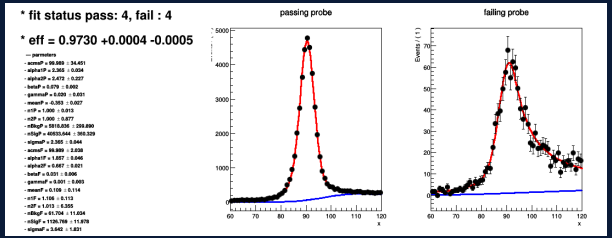
with NF



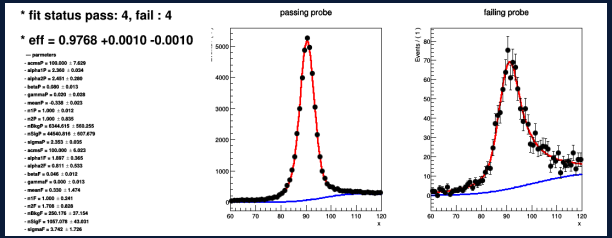
◀ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $70 < E_T < 90$ GeV

without NF



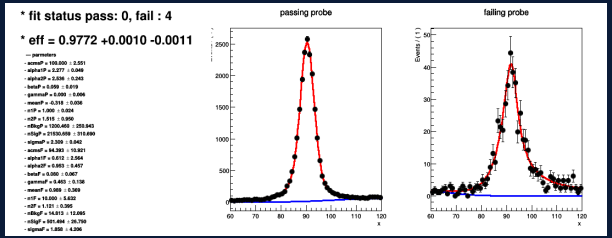
with NF



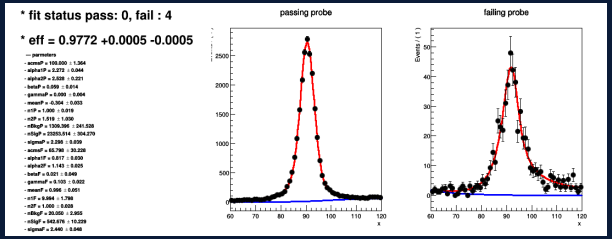
◂ efficiency

Fits: unseeded leg, EB $0.55 < R_9 < 0.85$, $E_T > 90$ GeV

without NF



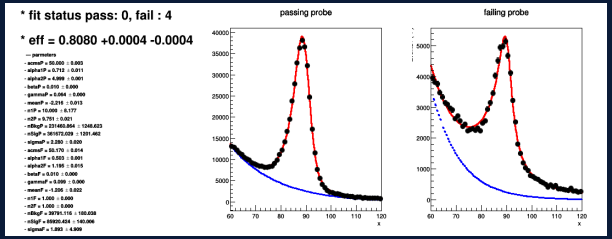
with NF



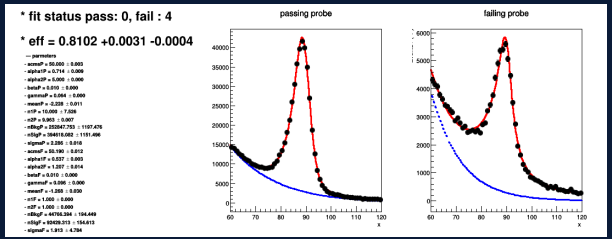
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $0 < E_T < 21$ GeV

without NF



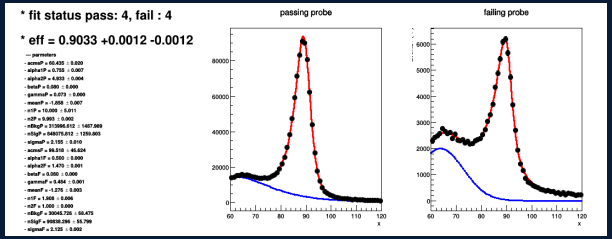
with NF



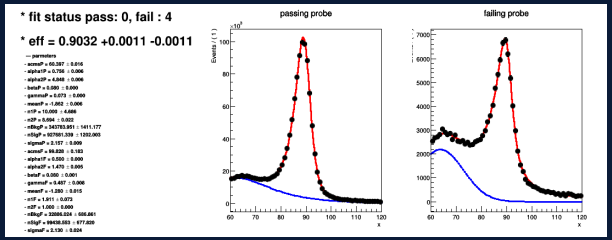
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $21 < E_T < 25$ GeV

without NF



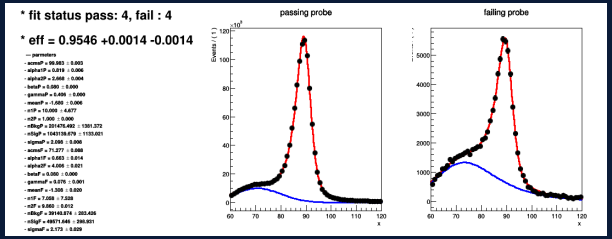
with NF



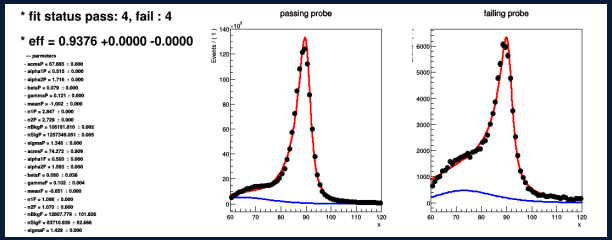
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $25 < E_T < 28$ GeV

without NF



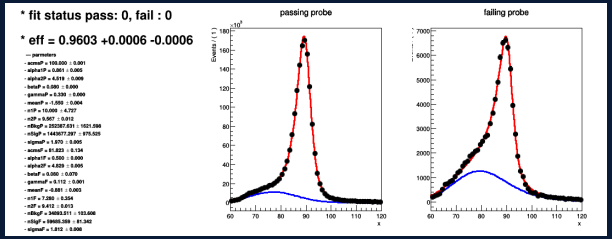
with NF



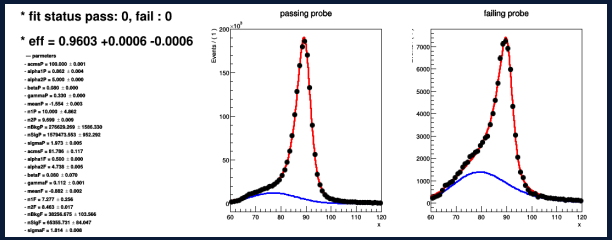
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $28 < E_T < 31$ GeV

without NF



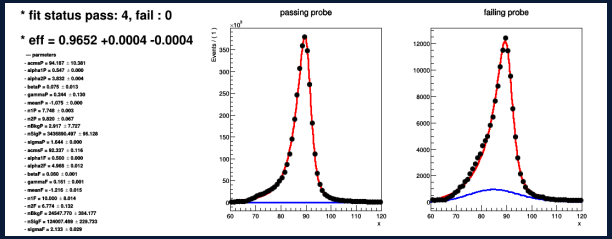
with NF



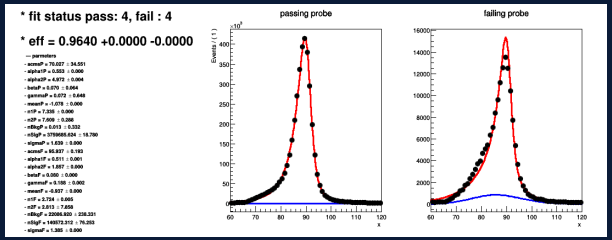
◀ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $31 < E_T < 35$ GeV

without NF



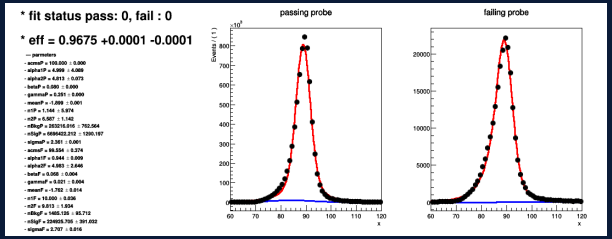
with NF



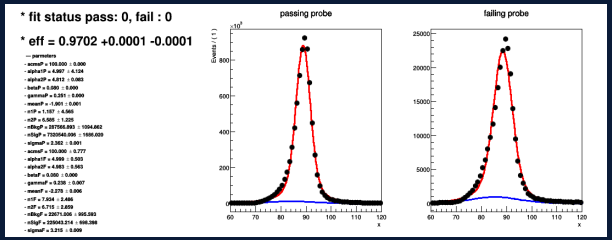
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $35 < E_T < 40$ GeV

without NF



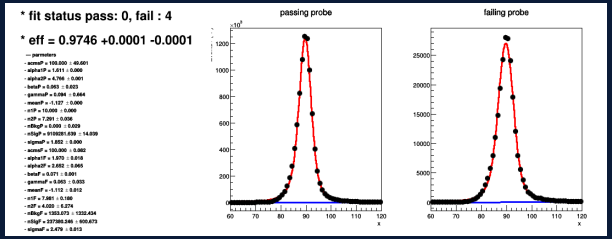
with NF



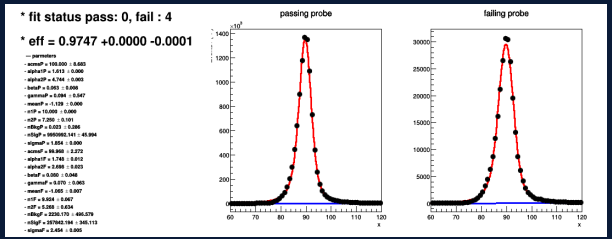
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $40 < E_T < 45$ GeV

without NF



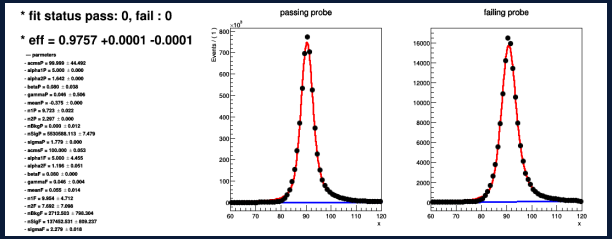
with NF



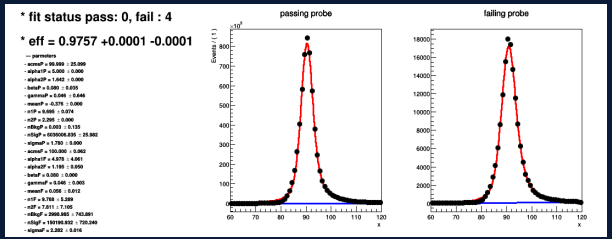
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $45 < E_T < 50$ GeV

without NF



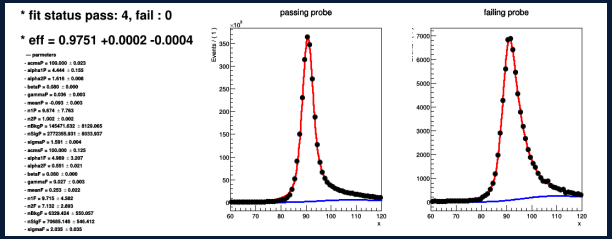
with NF



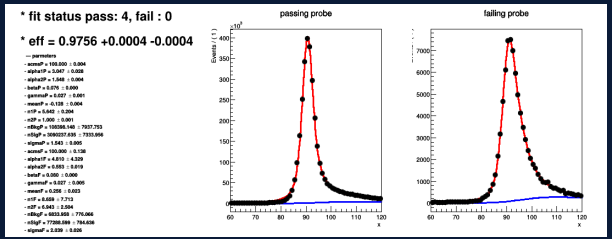
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $50 < E_T < 60$ GeV

without NF



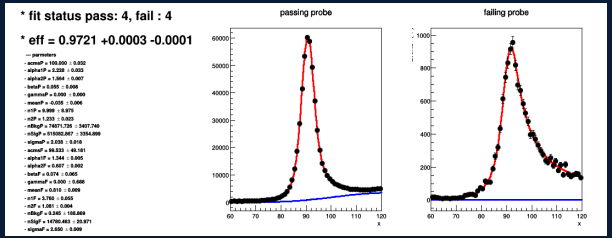
with NF



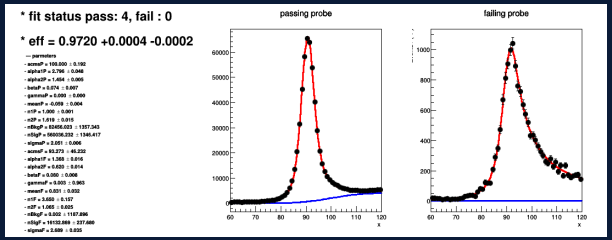
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $60 < E_T < 70$ GeV

without NF



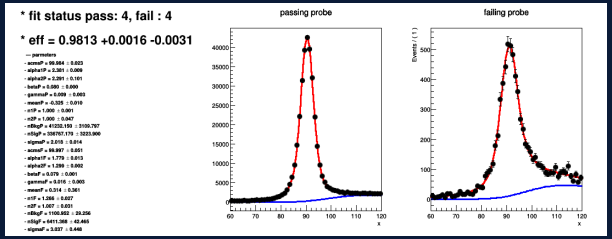
with NF



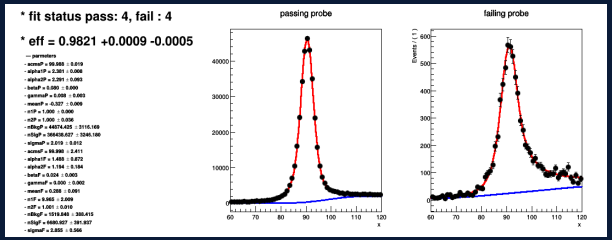
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $70 < E_T < 90$ GeV

without NF



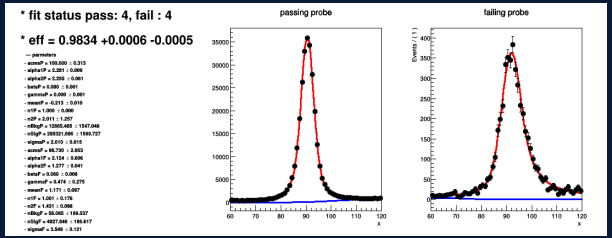
with NF



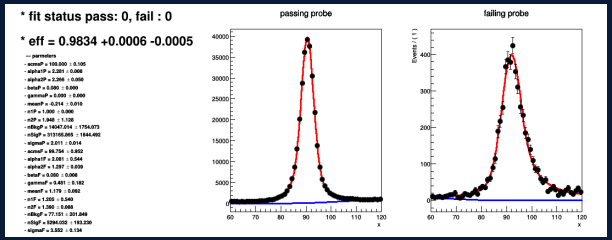
◂ efficiency

Fits: unseeded leg, EB $R_9 > 0.85$, $E_T > 90$ GeV

without NF



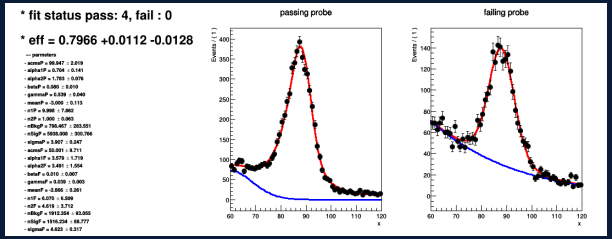
with NF



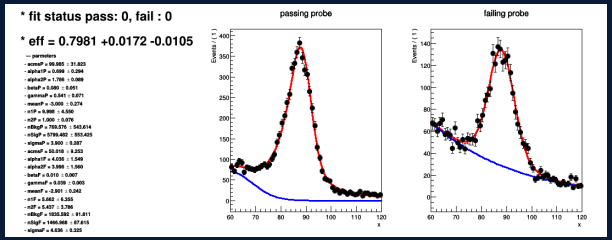
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $0 < E_T < 21$ GeV

without NF



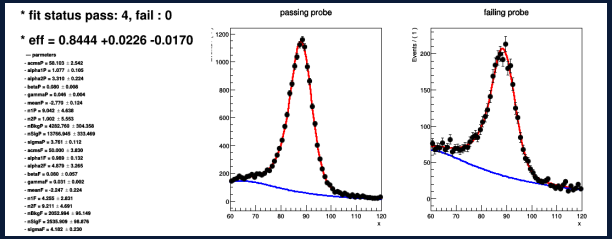
with NF



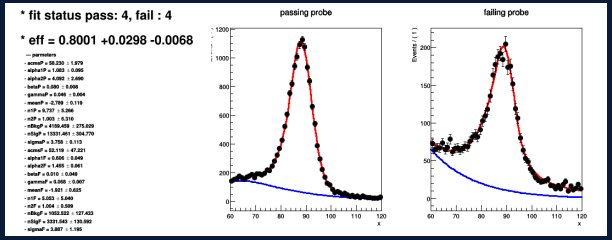
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $21 < E_T < 25$ GeV

without NF



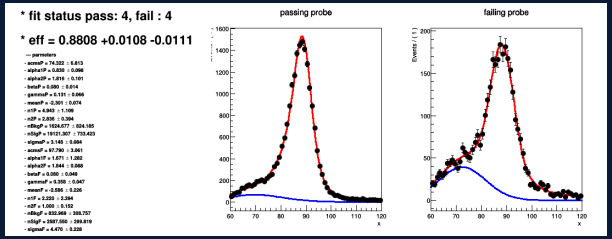
with NF



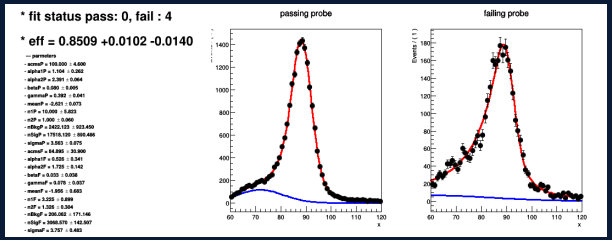
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $25 < E_T < 28$ GeV

without NF



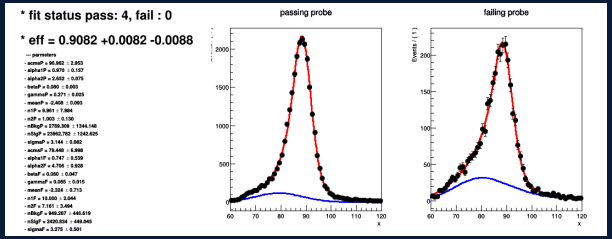
with NF



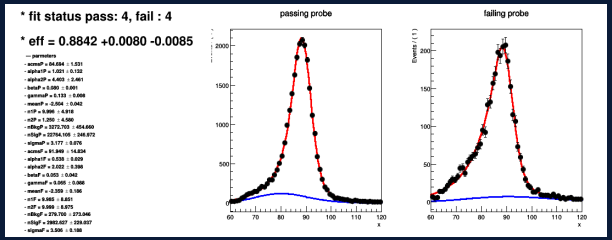
◀ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $28 < E_T < 31$ GeV

without NF



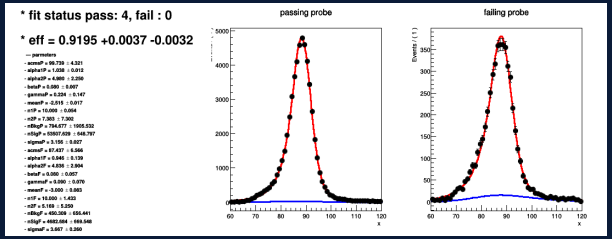
with NF



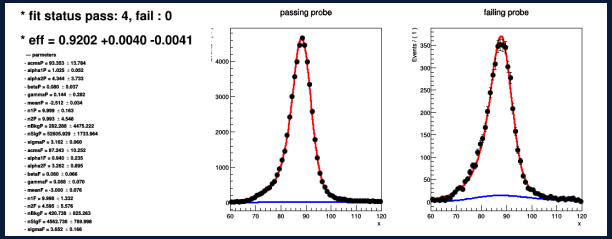
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $31 < E_T < 35$ GeV

without NF



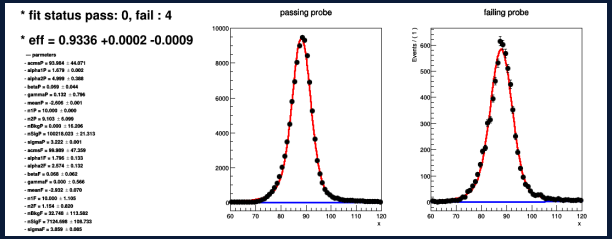
with NF



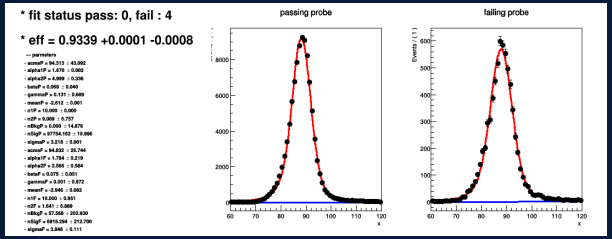
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $35 < E_T < 40$ GeV

without NF



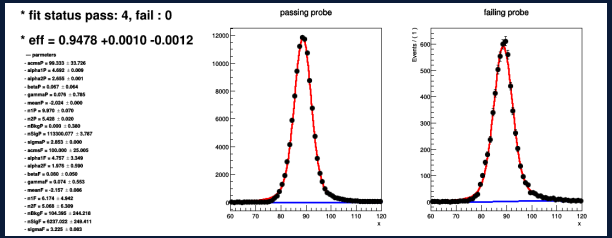
with NF



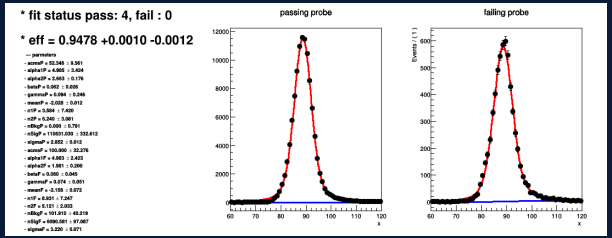
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $40 < E_T < 45$ GeV

without NF



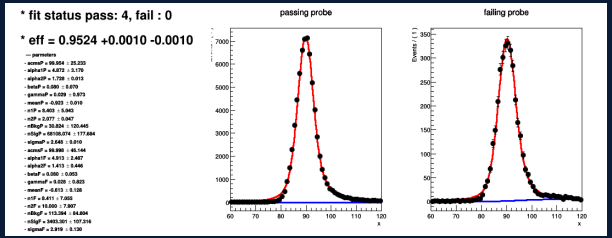
with NF



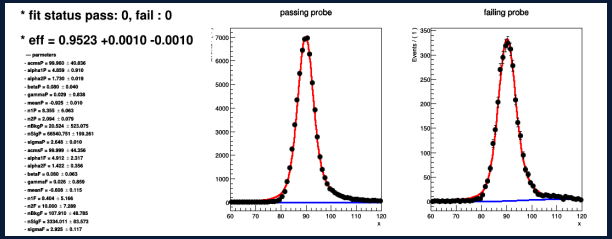
efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $45 < E_T < 50$ GeV

without NF



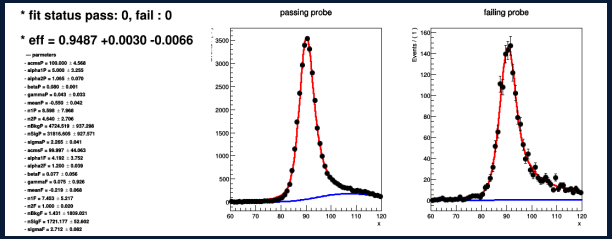
with NF



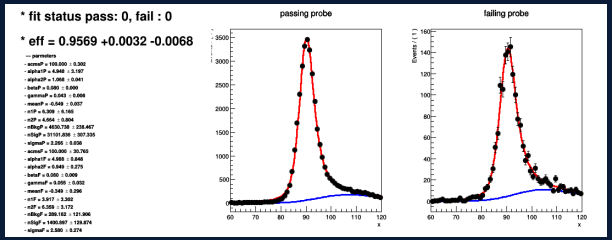
◀ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $50 < E_T < 60$ GeV

without NF



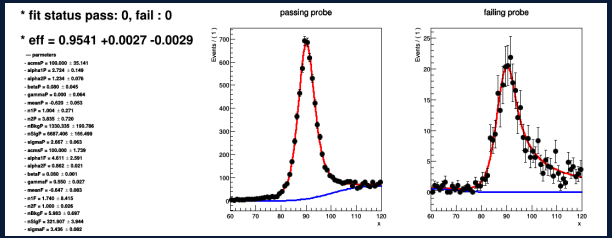
with NF



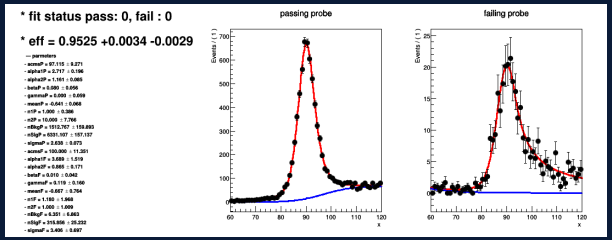
◀ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $60 < E_T < 70$ GeV

without NF



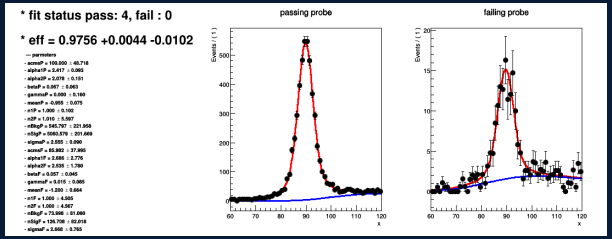
with NF



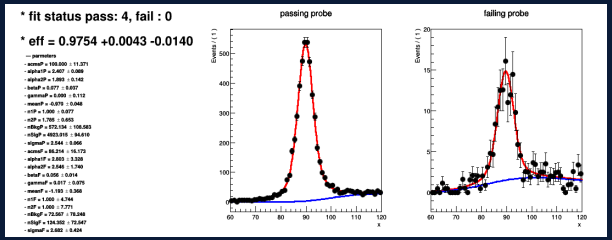
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $70 < E_T < 90$ GeV

without NF



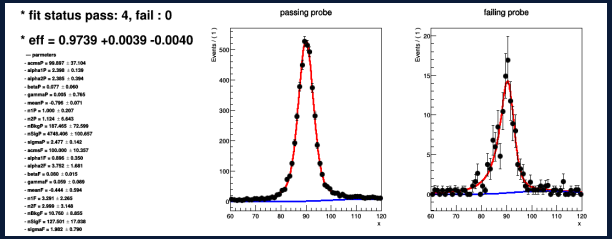
with NF



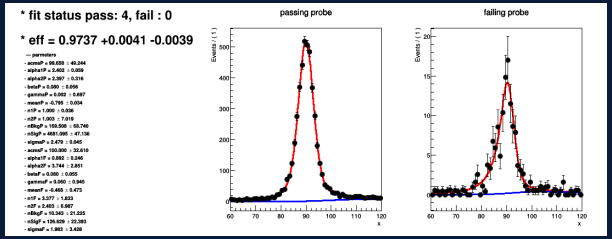
◂ efficiency

Fits: unseeded leg, EE $0.90 < R_9 < 0.93$, $E_T > 90$ GeV

without NF



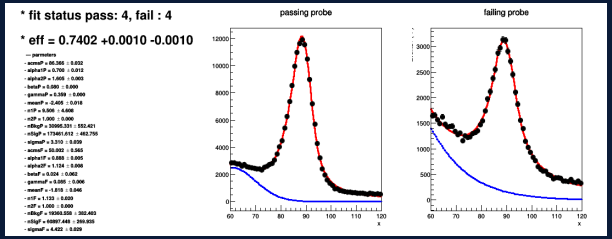
with NF



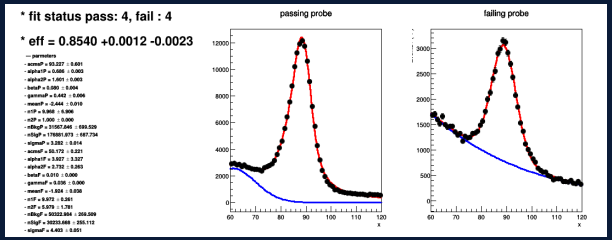
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $0 < E_T < 21$ GeV

without NF



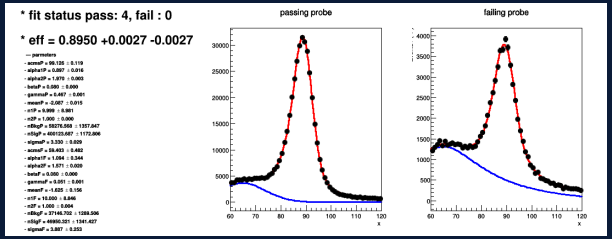
with NF



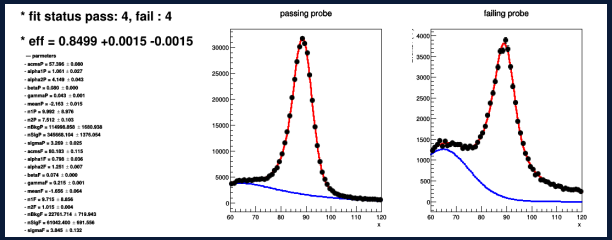
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $21 < E_T < 25$ GeV

without NF



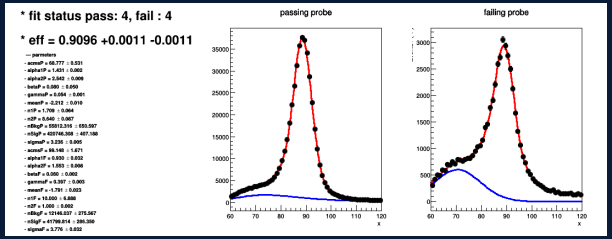
with NF



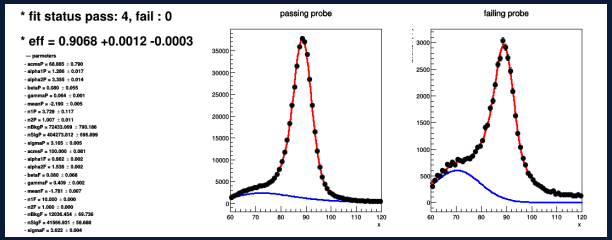
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $25 < E_T < 28$ GeV

without NF



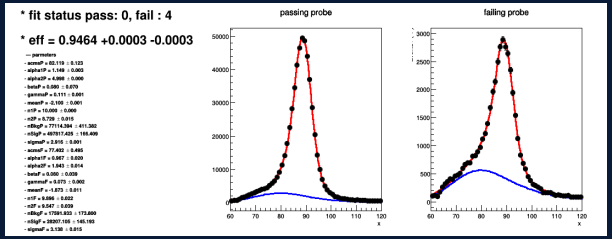
with NF



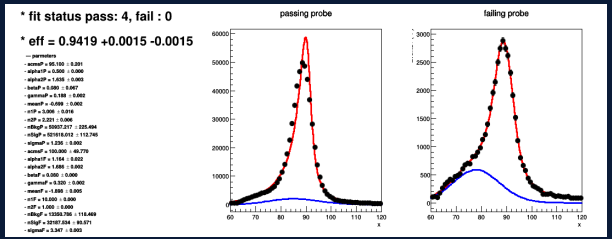
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $28 < E_T < 31$ GeV

without NF



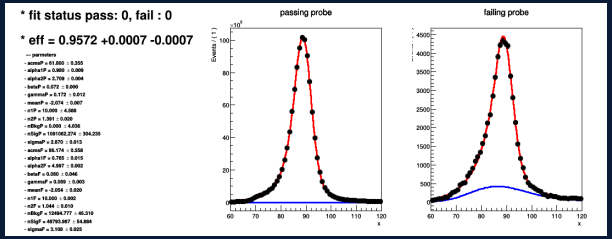
with NF



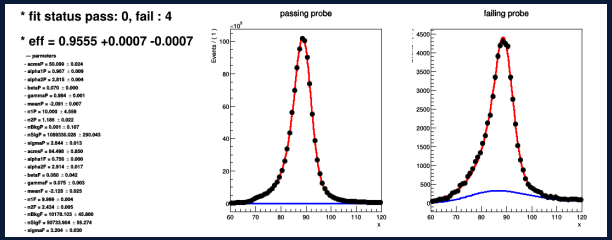
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $31 < E_T < 35$ GeV

without NF



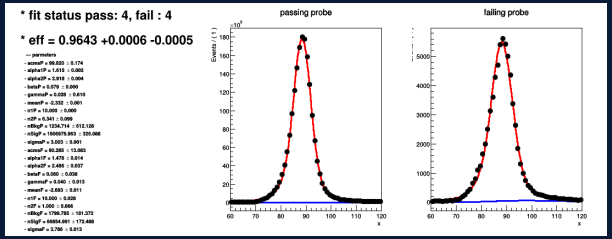
with NF



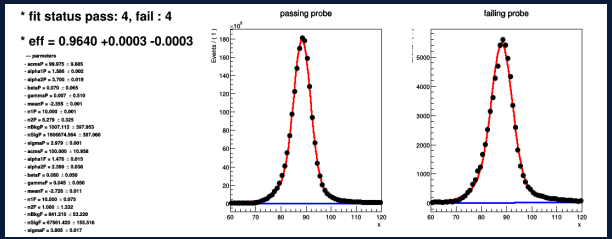
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $35 < E_T < 40$ GeV

without NF



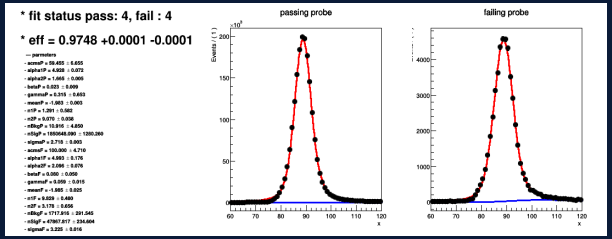
with NF



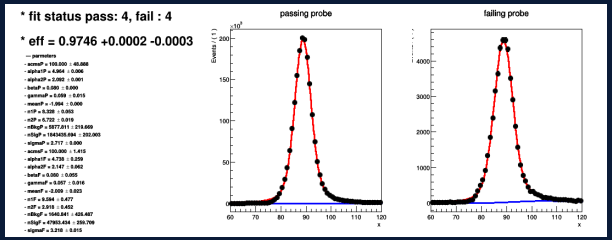
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $40 < E_T < 45$ GeV

without NF



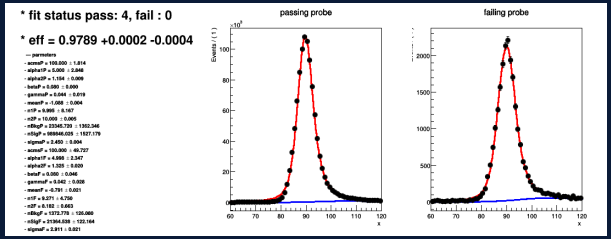
with NF



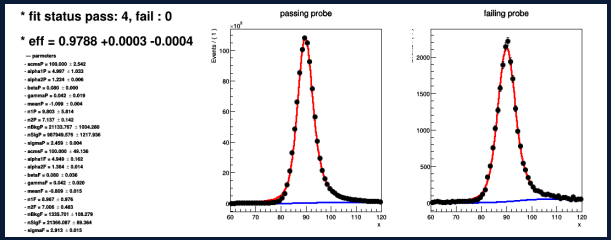
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $45 < E_T < 50$ GeV

without NF



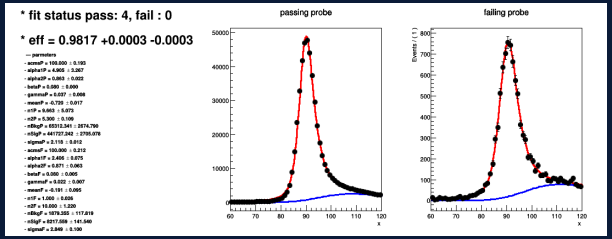
with NF



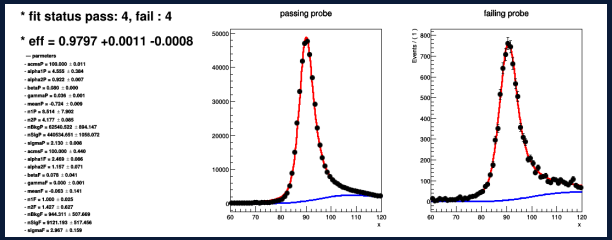
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $50 < E_T < 60$ GeV

without NF



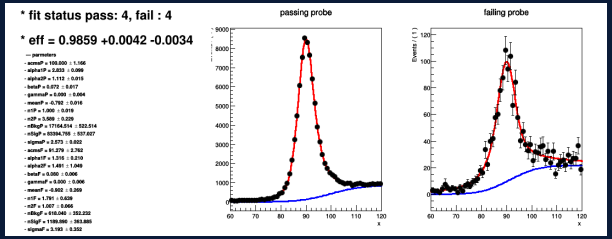
with NF



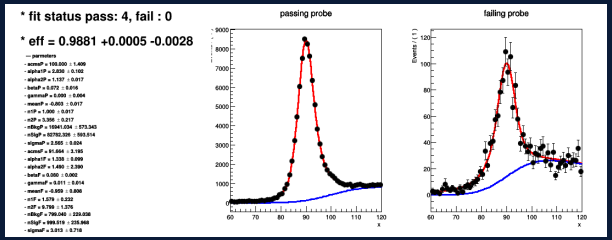
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $60 < E_T < 70$ GeV

without NF



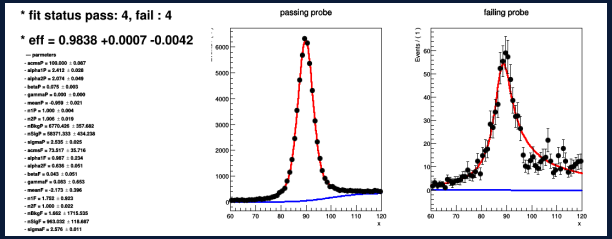
with NF



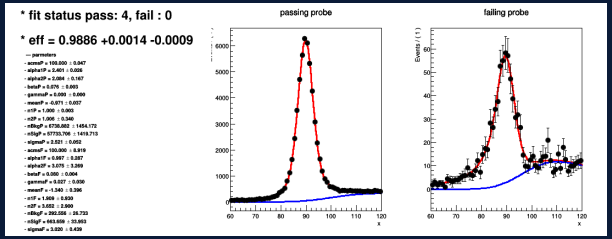
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $70 < E_T < 90$ GeV

without NF



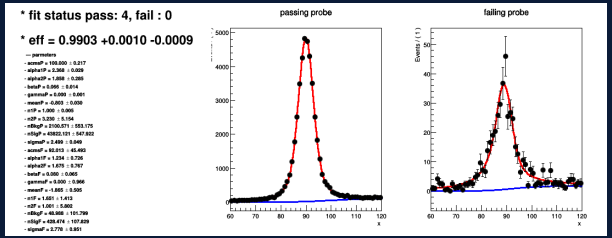
with NF



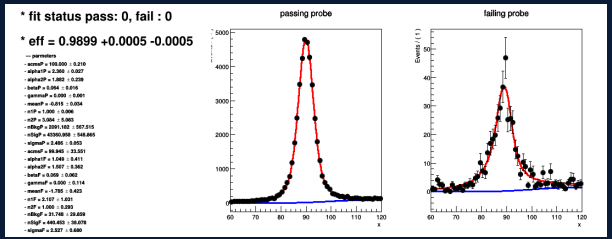
◂ efficiency

Fits: unseeded leg, EE $R_9 > 0.93$, $E_T > 90$ GeV

without NF



with NF



◂ efficiency